

The Effects of Education on Corruption: Evidence from Vietnam's University Expansion*

Edmund Malesky[†]

Duke University

Martin Mattsson[‡]

National University of Singapore

Khoa Vu[§]

Charles River Associates

Liaoliang Zhang[¶]

National University of Singapore

May 23, 2026

Abstract

Education and corruption are negatively correlated at the cross-national level, but little is known about the causal relationship between the two. We combine newly-collected data on timing and locations of Vietnam's national expansion of universities with detailed survey data on experiences of corruption from over 170,000 respondents in 320 districts across 12 years. Using staggered difference-in-differences, we show that cohorts exposed to the university expansion are 78% more likely to have a university education. However, this increase neither translates into fewer individuals being affected by corruption nor increases the propensity to denounce corrupt officials. Instead, we find that education increases exposure to corruption at the individual level. The mechanism for this increase that is most consistent with our data is that education raises household income and higher income leads to more bribe payments.

Keywords: Corruption, Education, Institutions and Development, Government Accountability

JEL Codes: D73, H75, I23, I25

*We thank Quoc-Anh Do, Ruben Durante, Lorenz Goette, Siddharth George, Jessica Pan, conference and seminar participants at the 6th World Labor Conference, the AASLE Conference, the IRLE Visiting Scholars Workshop, and the NUS SIG seminar for helpful comments and discussions. We thank Van Minh Le for his excellent research assistance. This project was supported financially by the National University of Singapore.

[†]ejm5@duke.edu

[‡]Corresponding author: martin.mattsson@nus.edu.sg

[§]vukhoa180692@gmail.com

[¶]liaoliang.zhang@u.nus.edu

1 Introduction

Corruption is a major global problem, but it is very unevenly distributed ([Andvig and Moene, 1990](#); [Svensson, 2005](#)). Corruption is ubiquitous in some countries, while requests for bribes and other forms of petty corruption are almost completely rooted out in other settings. Typically, the development of the economy and state capacity of nations goes hand in hand with decreases in corruption. However, we know very little about the causal mechanisms of this development process.

One potential reason why corruption decreases at higher levels of economic development is education, which typically increases substantially as part of the overall development process. There are multiple mechanisms through which education may reduce corruption. First, as more educated individuals know their rights and can refuse to make illicit payments when government officials demand them. Education may also increase the ability to comply with regulations and administrative processes, avoiding the need to pay bribes to bypass regulations. Second, educated citizens have more political and economic resources to hold malfeasant elites accountable, for example, by reporting corrupt officials ([Glaeser et al., 2007](#); [Botero et al., 2013](#)). Third, education may instill social norms that paying bribes is morally wrong.¹ Finally, education has been shown to motivate political participation and demands for better governance, which could lead to institutional and policy reforms that reduce opportunities for corruption and facilitate the selection of less corrupt officials ([Dee, 2004](#); [Milligan et al., 2004](#); [Wantchekon et al., 2015](#); [Larreguy and Marshall, 2017](#)).

There is also a strong positive cross-national correlation between education and the control of corruption, as shown in Appendix Figure [A.1](#). It is therefore natural to

¹For example, the prominent anti-corruption campaigner and founder of Transparency International, Peter Eigen, stated that “There are indeed moral dimensions to fighting corruption. It requires courage, honesty and integrity. Education is key in building such ethics in our society.”

hypothesize that expanding education could have the additional benefit of reducing corruption. In his influential review paper, [Svensson \(2005\)](#) cites [Lipset \(1960\)](#) and [Glaeser et al. \(2004\)](#), arguing that in their view “education and human capital is needed for courts and other formal institutions to operate efficiently, and government abuses are more likely to go unnoticed and unchallenged when the electorate is not literate.”

We test the hypothesis that expanding education reduces corruption by using the expansion of universities in Vietnam. Among low- and middle-income countries, Vietnam is considered highly successful in terms of educating its population ([Asadullah et al., 2020](#); [Dang et al., 2023](#)). From 2006 to 2013, the government expanded university education to a larger share of the population by building 122 new universities. For the population of college-age or younger citizens living in a district where a university opened, this expansion led to a 8.8 percentage point (78%) increase in the share of the population attending university. We use this rapid increase to test if higher levels of education reduce corruption.

To measure corruption, we employ the survey data used to create the Viet Nam Provincial Governance and Public Administration Performance Index (PAPI). The PAPI survey data contains detailed responses about corruption experiences among a representative sample of individuals from both rural and urban areas. We compile and harmonize 12 years of survey data from more than 170,000 individuals, providing precise and representative estimates of corruption measured in a consistent way over time for most of our outcomes. In particular, the PAPI data allow us to measure corruption in three ways: self-reported experiences of any type of corruption, perceptions of both petty and grand corruption, and bribe payments when interacting with government officials to apply for certificates and permits, as well as when accessing public services. We then linked these data with data on university opening dates and

locations compiled by [Vu \(2025\)](#).

We use an age-cohort difference-in-differences approach to estimate the effect of university openings on corruption.² We compare districts where a university opened with districts that did not have a university open, and we compare age cohorts that were of college-age or younger when the university opened, with age cohorts that were too old to have had their education directly affected by the university opening. Using this approach, we show that increasing the share of a cohort with a university education does not reduce the requests for bribes by government officials or subsequent payments. Instead, we show that both experiences with and perceptions of corruption *increase* with being exposed to a university opening, while our measure of actual payments when applying for permits or public services is unaffected. This suggests that at the individual level, more education leads to being more exposed to corruption. Given that university openings were not random, we test for violations of the parallel trends assumption using [Rambachan and Roth \(2023\)](#)’s partial identification approach, which suggests that our main results are not driven by pre-trends.

Our main results are at the individual- or cohort-level, and do not answer the question of how educating a society *as a whole* affects the overall level of corruption. We complement our main analysis using a standard difference-in-differences analysis to measure the effect on local communities, not just the cohorts directly affected by the increased access to education. We find no effect of educational expansion on any of our corruption measures.³ This suggests that from a policy perspective, expanding tertiary educational access does not alter corruption exposure, perception, or the behaviors of paying informal charges in the short- or medium-run.

²Similar to [Vu \(2025\)](#), we apply the staggered timing difference-in-differences to age cohorts as new universities were established in different years for different locations. We use a stacked regression approach to account for treatment effect heterogeneity and negative weights in conventional TWFE estimators ([Cengiz et al., 2019](#)).

³We comment on the assumption and limitation of this exercise in Section [4.2.4](#).

We also directly test the hypothesis that more educated individuals are more willing to report official misconduct to the authorities and that these frequent complaints reduce corruption (Lipset, 1960; Botero et al., 2013). We measure the effect of exposure to university openings on the probability of reporting a government official demanding a bribe, both conditional on having experienced such a demand and unconditionally. We also use responses to a separate hypothetical question as an additional measure of reporting behavior. Across all measures, we find no evidence that university education increases the probability that individuals report corrupt government officials to the authorities.

To better understand the *increase* in experiences of bribery among cohorts exposed to the university expansion, we investigate several potential mechanisms. We find no evidence for the hypothesis that more educated individuals are more comfortable reporting instances of corruption in the survey. Furthermore, we do not find any evidence that the results are driven by more educated government officials demanding more bribes or that migration to the districts where universities opened is affecting our results.⁴

The mechanism most consistent with our results is that individuals with higher incomes pay more bribes, either because they are more willing to pay, they are exposed to more bribe-paying situations, or because they are targeted by government officials. We show that education increases household income and that households with higher incomes are more likely to pay bribes. In particular, we find that being exposed to a university opening increases the propensity to pay middlemen when interacting with government officials. While our survey questions do not capture all interactions with government officials and reveal no increase in direct informal payments, evidence on

⁴Vu (2025) also found that university openings do not affect either in-migration or out-migration, which is consistent with our finding.

payments to intermediaries for government procedures suggests that a higher opportunity cost of time for more educated and richer cohorts may account for part of the observed increase in corruption exposure.

Our paper contributes to three strands of literature. First, there is extensive literature studying corruption in educational institutions as well as the effects corruption has on learning (e.g. [Reinikka and Svensson, 2005, 2011](#); [Ferraz et al., 2012](#)), but only limited evidence on how education affects corruption. Early empirical work by [Glaeser and Saks \(2006\)](#) uses historical congregationalism in US states as an instrument for the level of schooling today and finds that more educated states have less corruption as measured by the number of government officials convicted. Apart from that study, to the best of our knowledge, this is the first paper studying the causal effect of education on corruption. Our evidence shows that a massive and generally successful expansion of tertiary education did not decrease corruption at the individual level. Furthermore, we show that one of the mechanisms suggested by the literature, that educated individuals are more likely to report corruption, does not materialize in our context. Our findings are consistent with the descriptive findings of [Mocan \(2008\)](#), who shows that education is positively associated with being asked for a bribe at the individual level in developing countries.

Second, we add to the extensive body of research on the determinants of corruption, a significant subject in Economics, Sociology, and Political Science (e.g. [Treisman, 2000, 2007](#); [Campante and Do, 2014](#); [Fisman and Golden, 2017](#)). By examining experiences of corruption among cohorts exposed to university expansion in the short and long run, we complement applied literature that often examines shorter-term changes in corruption in response to specific policy interventions (e.g. [Olken, 2007](#); [Banerjee et al., 2020](#); [Mattsson, 2025b](#)). This is important as we only expect corruption to slowly reach a new equilibrium, as multiple actors adjust their behavior to the

new dynamics and incentives ([Olken and Pande, 2012](#)). Moreover, we are measuring not just one intervention for one government function; we are measuring the effect of having a university degree on the overall level of corruption experienced, across a wide variety of government institutions.

Lastly, we contribute to a broader literature on the roles of education in long-run economic development, particularly in debates over modernization theory and the primacy of economic or political foundations for long-term economic growth ([Lipset, 1960](#); [Glaeser et al., 2004](#); [Acemoglu et al., 2005, 2009, 2019](#)). At the fundamental level, access to education drives the accumulation of human capital, directly contributing to production ([Lucas Jr, 1988](#)), innovation and ideas ([Romer, 1990](#)), and economic growth ([Hanushek and Woessmann, 2015](#); [Hendricks and Schoellman, 2023](#)).⁵ A more recent literature focuses on the effect of education on political institutions, in particular through its effect on political participation and the ability to hold government officials accountable ([Dee, 2004](#); [Milligan et al., 2004](#); [Sondheimer and Green, 2010](#); [Campante and Chor, 2012](#); [Botero et al., 2013](#); [Wantchekon et al., 2015](#); [Croke et al., 2016](#); [Larreguy and Marshall, 2017](#)). Our paper contributes to this literature by studying education’s role in determining individuals’ behavior in everyday interactions with government institutions, instead of participation in well-defined political processes such as voting in elections.

The rest of the paper is structured as follows. Section 2 describes the context and the data. Section 3 describes our empirical strategy and Section 4 presents our results. Section 5 discusses mechanisms. Section 6 concludes.

⁵See [Valero \(2021\)](#) for an extensive review of this literature.

2 Context and Data

2.1 Corruption in Vietnam

Vietnam has a substantial corruption problem and is ranked at the 46th percentile for Control of Corruption of the 2022 World Bank’s Worldwide Governance Indicators, meaning that it is only better than about half of the 214 countries on the list.⁶ This creates a challenging business environment. Bribery incidence (the percentage of firms experiencing at least one bribe payment request) is 31.1% in Vietnam, relative to an average of 17.6% in East Asia and the Pacific, and an average of 13.3% of all economies, according to the World Bank Enterprise Survey. The combination of high levels of corruption with relatively high quality data has generated a substantial quantitative literature on corruption in Vietnam (e.g., [Do et al., 2013](#); [Malesky et al., 2015](#); [Bai et al., 2019](#); [Sharma et al., 2021](#)).⁷

To combat corruption, the General Secretary of the Communist Party of Vietnam initiated the “Blazing Furnace” anti-corruption campaign in 2016. While earlier efforts to limit corruption have fizzled out without much effect ([Malesky et al., 2019](#)), the campaign started in 2016 brought more meaningful changes. Since 2016, over 200,000 party members, including two Presidents, the Speaker of the National Assembly, and several Politburo members, have been disciplined ([Giang, 2024](#)).⁸ Surveys of firms and citizens also indicate that petty corruption has decreased. For example, in 2016, 66% of firms reported that informal charges are common, while the same number

⁶See [World Bank \(2022\)](#).

⁷In particular, [Bai et al. \(2019\)](#) shows that bribery decreased over time as businesses became more productive and mobile, improving their ability to negotiate lower bribe schedules from local government officials. This work differs from ours in three important ways: 1) they study firm-level behavior from business surveys while we focus on citizens; 2) productivity improvements are modeled as industry-specific shocks rather than improvements in education; and 3) their measure of corruption is the share of bribes over sales revenue, while we focus on the frequency of different types of illegal payments.

⁸For more details on the anti-corruption campaign see Appendix [A.1](#).

in 2023 was only 33% (Malesky et al., 2024). Vietnamese citizens also report lower levels of bribery and are more likely to believe the government is willing to combat corrupt activities (PAPI, 2023). It is possible that the scale and intensity of this anti-corruption campaign moderated the impact of education on bribery, as it created more avenues for contesting and appealing corrupt activity. We explore this possibility in more detail in Section 5.3.

2.2 Vietnam’s University Expansion

Against the backdrop of rising demands for skilled labor in the early 2000s (McGuinness et al., 2021), in 2006, the government issued Decree 121/2007, which served both as a policy road-map for a national expansion of higher education and a green light for local governments and private enterprises to apply to open new universities. The decree laid out the government’s strategy of rapidly scaling up higher education and established an overall expansion plan for the 2006 to 2020 period.

The expansion resulted in the establishment of 122 new universities across the country from 2006 to 2013.⁹ As new universities were opened, the number of university and college teachers also rose substantially, from 48.6 thousand in 2005 to 91.4 thousand in 2014 (Vu and Nguyen, 2018). The number of tertiary-level students increased rapidly from 1.4 million to 2.1 million between 2005 and 2014, representing an increase in the gross enrollment rate from 16.1% to 30.4%.

Prior to the expansion in 2006, universities were highly concentrated; only 62 out of 703 districts had a university. During the expansion, new universities were opened in 79 districts, 48 of which previously lacked a university (see Appendix Figure A.2).¹⁰

⁹This includes existing universities opening campuses in a new province. The district codes are standardized to the 2019 population census published by the General Statistics Office (GSO) of Vietnam (Ruggles et al., 2025).

¹⁰Based on district-level population counts, the university expansion directly affects 14.5% of Vietnam’s population in the same district (Ruggles et al., 2025).

We use the establishment of these universities to estimate the effect of a new university being established on both education and corruption.

It is important to note that new university locations are not random. New universities may be opened in response to rising demand for skilled labor or conditions that make establishing a university feasible, causing them to be in locations that are systematically different from those without university openings. In Appendix Table A.1, we report some differences in district characteristics between districts with and without university openings. For example, although the distances to special class cities are similar,¹¹ districts with university openings are more likely to be urban and provincial capitals, with smaller geographical size, higher population density, and a larger share of the population being the ethnic majority.¹² Such factors are only concerning to our identification strategy to the extent that they also affect corruption differentially across different birth cohorts conditional on district fixed effects and birth cohort fixed effects. However, there is little reason to believe that market-level economic factors would affect the corruption experiences of one birth cohort differently than those of another cohort, and that these differential effects would vary systematically across districts with and without university openings. We further discuss non-random university location concerns in Section 3.1.

2.3 Data on Education and Corruption

To measure education and corruption, we use the Vietnamese Provincial Governance and Public Administration Performance Index (PAPI) annual survey from 2011 to 2022. Our sample consists of more than 170,000 surveys from repeated cross-sections

¹¹Only Ho Chi Minh City and Hanoi are special class cities with the largest populations.

¹²Vu (2025) shows that new universities are often located in provinces with more manufacturing and service share of workers, which tend to be more skill-intensive than the agriculture sector. Vu (2025) also shows that provinces that opened a university and provinces that did not open a university during the expansion tend not to differ along any other observable dimensions.

of randomly selected individuals living across 320 districts.¹³ For more details on the PAPI survey, see Appendix A.2.

Education is measured with a variable that indicates whether the respondent has at least some university education.¹⁴ We do not classify vocational training programs or college courses as university education.

Corruption is notoriously challenging to measure (Olken and Pande, 2012), but the PAPI survey employs several methods to create a comprehensive set of corruption measures. In particular, we use three different types of corruption measures from the survey. First, we use self-reported experiences of corruption. We create two indicator variables “personally affected” and “household affected” based on whether the respondent states that they themselves were asked for a bribe by a government official in the past year, or if a household member was asked.¹⁵ As this survey question does not directly ask the respondent if they paid the requested bribe or not, we interpret it as indicative of being affected by or exposed to corruption, rather than having paid a bribe. We use these as our main measures of corruption as they are the most comprehensive measures of corruption experiences across all interactions with government officials.

¹³Districts are selected using probability proportion to size sampling. These districts cover 46.5% of Vietnam’s population based on the 2019 population census published by the General Statistics Office (GSO) of Vietnam (Ruggles et al., 2025).

¹⁴In a robustness test, we show that our results are similar when restricting our measure of education to completed university degrees.

¹⁵The question asked is “In the past year, have you or a family member ever been asked for a bribe or solicited for illicit payments by a local official or civil servant? Please consider all forms of bribery, ranging from small acts such as paying unofficial fees to a traffic police officer, to larger acts such as paying commissions to officials or civil servants in order to obtain a government procurement contract. [Trong một năm qua, ông/bà hay người thân trong gia đình có lúc nào bị một cán bộ, công chức ở địa phương vòi vĩnh đòi hối lộ hay chưa? Xin ông/bà nghĩ tới tất cả các dạng hối lộ, từ những việc nhỏ như trả tiền ngoài quy định cho một cảnh sát giao thông tới những việc lớn như trả hoa hồng cho cán bộ, công chức để có được hợp đồng thầu với nhà nước.]” The respondents could answer using the options: “not affected,” “personally extorted,” [Cá nhân tôi bị vòi vĩnh] or “other family members extorted” [Người thân bị vòi vĩnh]. The variable “household affected” indicates whether either of the two answers is yes.

Second, we use corruption perceptions. In particular, the PAPI survey includes questions about perceptions of the need to pay bribes to get land use rights, government jobs, public medical services, and more attention paid to the child in school. The survey also includes questions about perceptions of government officials diverting government funds for their personal benefit and getting “kickbacks” for the approval of construction permits. While corruption perceptions have been shown to be predictive of actual corruption, they are also known to be a noisy proxy (Olken, 2009). A benefit of perception measures is that respondents who may not be comfortable discussing their own experiences with corruption may be more willing to describe their general perceptions.

Finally, we use self-reported payments of bribes or payments to middlemen during six specific types of government interactions.¹⁶ Four of these interactions are government procedures: applying for certifications, construction permits, or land use rights, as well as “other administrative procedures” such as applying for state subsidies. The other two interactions occur when accessing public services, in particular public primary schools and public hospitals. Using self-reported experiences of corruption during these six interactions with government officials has the benefit of allowing us to separate out corruption experiences conditional on having an interaction with the government from corruption experiences for all respondents regardless of their interactions with the government. Furthermore, for three of the interactions, list experiments were used to measure the frequency of corruption experiences. The list experiments are designed to elicit information about corruption experiences without the respondent having to admit to paying a bribe and may therefore overcome the

¹⁶While payments to middlemen are not necessarily indicative of corruption, paying a middleman is a common way to pay a bribe. Therefore, we include these payments as bribe payments in our data. Appendix Table A.4 shows that excluding payments to middlemen yields results that are similar to our main results.

problem of social desirability bias or non-response due to fears of government reprisals (Malesky et al., 2015; Agerberg, 2022).

While none of these measures is perfect, we believe that together they provide some of the most detailed and comprehensive self-reported data on corruption available. As each measure has its own benefits and drawbacks, we present results for each of them separately. For more details on the construction of the outcome variables, including the exact phrasing of the questions, see Appendix A.5.

3 Empirical Strategy

We apply the empirical strategy proposed by Vu (2025) to study the effects of new university establishment on individual-level outcomes by leveraging the variation in the opening dates and locations of universities to identify the effects of a university being established on individuals. Specifically, we compare districts with universities established for the first time (treatment districts) and those that never had a university before (control districts) across different age cohorts.¹⁷

We define the age cohorts who were 23 or older when a new university opened as the unexposed cohorts and those who were 22 or younger as the exposed cohorts. This is an appropriate age cutoff in the context of Vietnam because individuals aged between 18 and 22 can (and are encouraged to) transfer from vocational and community colleges that typically provide two and three-year programs to a four-year university program and, thus, can still benefit from a new university opening. Consistent with the results of Vu (2025), we also show in Figure 1 that the expansion increases the college education rate for these “partially exposed” age cohorts.

¹⁷Vu (2025) uses variation of timing and location of university openings at the province level instead of the district level to capture the equilibrium effects of local labor markets at the province level, which is unrelated to this study. Therefore, we leverage district-level variation, which provides a more granular level of treatment.

Our natural starting point is a two-way fixed effects (TWFE) approach. Denote d and g the district and the year d had its first university, t the survey year, and c the age cohort relative to the first university opening year. A standard TWFE event study takes the following form:

$$y_{i,d,c} = \sum_c \delta_c * \mathbf{1}(Cohort_{i,d,c} = c) + \gamma_d + \eta_a + \theta_t + \epsilon_{i,d,c} \quad (1)$$

where $\mathbf{1}(Cohort_{i,d,c} = c)$ denotes an indicator for whether individual i belongs to cohort c relative to the university opening year, while γ_d , η_a , and θ_t denote district, age, and survey year fixed effects. The coefficients δ_c capture the treatment effect for cohort c . For individual-level outcomes, such as education or corruption perception, we omit the age cohort 23–27 as the reference group. We control for gender and ethnicity, and in certain robustness specifications, we additionally control for migrant status, party membership, and a government-official indicator.

For household-level outcomes, we instead use the age cohort 28–32 as the reference group. This accounts for the fact that the age cohort 23–27 might face potential spillover effects from younger family members who are exposed to the university expansion.¹⁸ This is a particular concern in Vietnam, given that young Vietnamese adults tend to live in the same household with someone from a younger age group, such as younger siblings or cousins. According to the Population and Housing Census 2009, around 30 to 35% individuals aged 20–30 live with someone from an immediately younger age group.

This baseline approach is prone to two well-known issues in the difference-in-differences literature. First, it depends on the parallel trends assumption, which may potentially be violated when new university locations are not random. Sec-

¹⁸In Section 4.3, we show that our main results are robust to using the same 28–32 age bin as the reference group for all main outcomes.

ond, the TWFE estimator may suffer from biases in a staggered timing difference-in-differences design when treatment effects are heterogeneous (De Chaisemartin and d’Haultfoeuille, 2020; Goodman-Bacon, 2021; Roth et al., 2023). We discuss how to address these concerns for the rest of this section.

3.1 Parallel Trends

The non-random nature of university locations poses a threat to the parallel trends assumption if the determinants of university locations also affect corruption trends. We assess this potential violation in two ways. First, we conduct conventional pre-trend tests to evaluate the counterfactual parallel trend assumption. Because older age cohorts not exposed to university expansion form the basis of these tests, the absence of a pretrend implies that parallel trend violation would require location-based confounders to affect corruption trends for younger cohorts but not for older cohorts, which is a more restrictive criterion for potential confounders. For robustness, we also conduct pre-trend tests for different regions across Vietnam.

Second, we re-estimate our results based on potential pre-trend violation. Specifically, we account for linear extrapolation of the pre-trends into the post-treatment period using Rambachan and Roth (2023)’s smoothness restriction test. Rambachan and Roth (2023)’s approach uses partial identification to estimate the confidence interval for the treatment effect when the violation of the parallel trends in the post-treatment period (which is unobserved) is less than or equal to a fraction ($\bar{M}$) of the largest deviation from the parallel trend in the pre-treatment period. For example, the confidence interval for $\bar{M} = 1$ includes the true treatment effect at least 95% of the time when the violation of parallel trend in post-treatment is at most equal to the largest deviation in the pre-treatment period.

Formally, denote $\gamma_{t=\{pre,post\}}$ difference in trends between the control and treatment groups (i.e., pre-trends and post-trends). The parallel trends assumption is $\gamma_{post} = 0$, which is unobserved, and γ_{pre} can be estimated in an event study regression. [Rambachan and Roth \(2023\)](#)’s approach yields a confidence interval of treatment effect when post-trends are only as large as $\bar{M}$ of the largest pre-trends period:

$$|\gamma_{t+1} - \gamma_t| \leq \bar{M} \cdot \max_{1 \leq k \leq t} |\gamma_{k+1} - \gamma_k|$$

where k and t denote the pre and post periods. The robust confidence intervals account for the fact that there could be estimation error both in the treatment effects estimates and estimates of the pre-trends.

3.2 Biased TWFE Estimate when Treatment is Staggered

Another concern with the use of TWFE approach with a staggered timing design is that potential heterogeneous treatment effects can bias our TWFE estimator ([Goodman-Bacon, 2021](#); [Callaway and Sant’Anna, 2021](#)). Specifically, the TWFE approach yields a weighted average treatment effect of the underlying individual treatment effects. As pointed out by [De Chaisemartin and d’Haultfoeuille \(2020\)](#), [Goodman-Bacon \(2021\)](#) and others in this extensive literature, when treatment effects are heterogeneous, this estimator might be biased by negative weights; in extreme cases, negative weights may lead to an overall treatment effect with an opposite sign from all individual treatment effects ([De Chaisemartin and d’Haultfoeuille, 2020](#)). The specific component of the TWFE estimator driving such negative weights is the comparison of the not-yet-treated group (as treatment) to the already-treated group (as control); negative weights arise when the treatment effect on the control group (already-treated) evolves over time, contaminating the treatment effect on the not-yet-treated group

(De Chaisemartin and d’Haultfoeuille, 2020; Goodman-Bacon, 2021).

We adopt Cengiz et al. (2019)’s stacked regression approach to alleviate this concern. This approach compares each treatment group with a clean control group (i.e., never treated) and aggregates across these comparisons while controlling for comparison-specific fixed effects, thus directly avoiding the “forbidden” comparison between a not-yet treated group and an already-treated group in the TWFE approach. We modify this estimator to accommodate an age-cohort difference-in-differences design, in which cross-sectional individuals observed in the same survey year face differential exposure to university openings across age cohorts. This contrasts with conventional staggered difference-in-differences estimators that define treatment variation along a single time dimension (e.g., years since treatment).¹⁹

Formally, the stacked design creates sub-datasets with only districts in G_g (treatment group with university opening year g) and the never-treated districts (control group):

$$y_{i,d,c}^G = \sum_c \delta_c^G * (T_{i,d}^G * Cohort_{i,d,c}^G) + \gamma_d + \eta_a + \theta_t + \epsilon_{i,d,c} \quad (2)$$

where $y_{i,d,c}^G$ denotes the outcome of individual i in district d of cohort c ; $T_{i,d}^G$ indicates whether district d had the first university during the expansion; $Cohort_{i,d,c}^G$ indicates the age cohort of individual i . District and age fixed effects are γ_d and η_a for this specific sub-dataset with university opening year g . $\delta^{G,t}$ captures the treatment effect of one specific sub-dataset. One can then aggregate δ^G across all G to obtain the weighted average effect for different groups of districts.

Therefore, alternatively, we can combine all sub-datasets s and estimate the fol-

¹⁹Conceptually, we could define age since treatment to combine the survey year and birth cohort dimensions; it is not straightforward to operationalize or interpret existing staggered difference-in-differences estimators due to sample size issues arising from the long age window before and after the university opening, as well as age and survey year fixed effects.

lowing model:

$$y_{i,d,c,s} = \sum_c \delta_c * (T_{i,d,s} * Cohort_{i,d,c,s}) + \gamma_{d,s} + \eta_{a,s} + \theta_{t,s} + \epsilon_{i,d,c,s} \quad (3)$$

where the fixed effects further interact with the sub-dataset label. This approach, proposed by [Cengiz et al. \(2019\)](#), enables a clean comparison between treatment group G with a never-treated control group in each given cohort, thus circumventing the problem of negative weights due to treatment effect heterogeneity.

4 Results

4.1 Effect of University Openings on Education

In Figure 1, we present age-cohort difference-in-differences estimates by using the specification in Equation 1. We estimate the effect of a university opening on the share of the population with at least some university education. The impact on university education is large and positive. When comparing cohort-level differences in education level across districts that established universities and those that never had a university, we observe in Table 1 that the share of the younger cohorts with a university education increases by 8.8 percentage points or 78% for those aged 22 and below when the university opened, relative to the mean of districts that never had a university.²⁰ Our main university education measure includes individuals who received some university education but did not graduate. As a robustness test, we also estimate the effect on the share of the population with a completed university

²⁰We explain our choice of reference group in Section 3.

degree in Appendix Figure A.3, and the results are very similar.²¹

Our empirical strategy relies on an identifying assumption that in the absence of the university expansion, cohort-level differences would trend similarly across the treated districts that have a university opened, and control districts that never have a university opened. In other words, there are no omitted time-varying and district-specific confounders that are correlated with university expansion. Although we cannot test this assumption directly, we show that for the older cohorts who were 23 and above when the university opened, there are no differential pre-trends. Any omitted confounders would therefore have to affect only younger cohorts aged 22 and below.

4.2 Effect of University Openings on Corruption

4.2.1 Effect on Self-reported Experiences of Corruption

Figure 2 presents our main result for the effect of exposure to a university opening on the incidence of corruption at the individual level. We apply the same empirical framework from Equation 1 and report the effects on two corruption measures, an indicator of personally having been extorted by a government official (shown in red with triangle markers) and an indicator that also includes a family member having been extorted (shown in blue with square markers). As explained in Section 3, we regard the age cohort from 28 to 32 as the reference group for this outcome to account for the potential spillover effects from younger family members.²² Table 2 shows corruption incidences increased significantly by 53.2% for younger cohorts aged at 27

²¹The share of the younger cohorts with a completed university degree increases by 7.0 percentage points or 66.8% for those aged 22 and below when the university opened. This is consistent with the results of Vu (2025) who used data from Vietnam’s Labor Force Survey to estimate that the university openings increased college completion by 57%.

²²An individual between 23 and 27 years old can have a younger sibling or a younger spouse, between 18 to 22 years old, who is exposed to university expansion, so the effects on the younger member’s education (shown in Figure 1) and households’ corruption exposure start to manifest. For transparency, Appendix Figure A.4 shows this pattern using the reference group of 23 to 27.

or below when the university opened, relative to the mean of districts that never had a university open. The sharp increase in corruption exposure also only occurs in the younger age cohorts exposed to university expansion, and there are no differential pre-trends among respondents who were too old to be exposed to the university opening.

One potential confounding factor is that survey respondents with more education may be more comfortable talking about corruption. If this were the case, we should see a decrease among survey respondents refusing to answer the question, or answering by saying “I don’t know.” Appendix Figure A.5 estimates the effect on respondents not answering a corruption question in these ways. Panel (a) shows the effect on the answering behavior for whether personally or the household is affected by bribe-taking. Panels (b) and (c) show the effect for the questions about informal payments during specific government interactions, conditional on having had the interaction with the government or for all respondents.²³ We do not find any evidence that being exposed to the university opening affected the share of such answers.

4.2.2 Effect on Perceptions of Corruption

We now turn to the effect of being exposed to a university opening on perceptions of corruption. Figure 3 and Table 3 show that new university openings increased perceptions of corruption across a wide range of government activities. Respondents were asked to what extent they agreed with different statements about corruption and we present the results as standard deviations in the responses. Respondents exposed to the university opening are more likely to agree with the statement that people have to bribe to obtain a land title, a construction permit, or to have better care

²³Households can have various ways of interacting with government procedures or services, and we will discuss specific interactions below. If a bribing question for any of the interactions is answered with “don’t know” or “refuse to answer,” the indicator variables have the value 1.

for their children in primary schools,²⁴ or to get a government job. They are also more likely to agree with the statement that “officials divert funds for their personal benefit.” While the effects on agreement with the statement that one has to bribe for “medical treatment in the district’s hospitals” is not significant, it is positive and consistent with the other results. We create an overall index that captures all these components. We detail the construction of the perception measures in Section A.5. Overall, exposure to a university increases this corrupt perception index by 0.06 standard deviations. We also test the robustness of this index in Appendix Table A.2 with additional controls, and the results are very similar.

4.2.3 Effect on Corruption in Specific Interactions with the Government

Finally, in Figure 4 we show the effect on corruption during specific interactions with the government, both conditionally on having these interactions and for all respondents. We construct two summary measures, one for making informal payments while applying for public certifications, construction permits, land use rights, and other administrative procedures, such as state subsidies. The other measure is based on making informal payments while accessing public services at hospitals and primary schools. We explain in detail how the measures are constructed in Section A.5.

In panels (a) and (b) of Figure 4, we combine both paying the middleman and bribes as informal charges and pool them across different government procedures. Panel (a) shows the results conditional on having had the interaction, while panel (b) shows the results for everyone in the population. We do not find evidence in support of changes in corruption as a result of the exposure of a university opening. In panels (c) and (d) of Figure 4, we estimate the effect on corruption in the two government

²⁴Educational bribes are usually paid to teachers in the form of payments for extra tutoring outside of class - with an implicit agreement that this will improve test performance.

services, conditionally and unconditionally on having obtained the government service. We do not observe any effect of education on individuals' bribing behaviors when obtaining these services. The point estimates of the effects of university expansion on these measures of corruption are reported in Appendix Table A.3. All point estimates are positive but close to zero and not statistically significant. We also examine informal payments for government procedures separately from payments to the middleman in Appendix Table A.4.²⁵ We find no empirical evidence of changes in bribing behavior.

To reduce concerns of under-reporting of corruption, the PAPI survey uses a list experiment when asking if a bribe was paid during an interaction with the government. This is in addition to the direct question used for the analysis in Figure 4. In the list experiment, the survey respondents are given a list of actions they may have taken during the interaction with the government. One action is corruption-related, while the other actions are not. Respondents are randomized into being shown a list of all actions or a list with the corruption-related action removed. Comparing responses from the group shown the corruption action and the group not shown the corruption action allows us to estimate how many people took the action on average, although we cannot know definitively which specific individual was asked for a bribe. In Appendix Table A.5, we estimate the share of people having taken the corruption action using the two-stage estimator developed by Blair and Imai (2012).²⁶ We find no evidence of a reduction in informal payments when using the responses from the

²⁵Questions on payments to middleman are only asked for government procedures and not government services.

²⁶Blair and Imai's list-experiment estimator uses a two-stage procedure in which the distribution of responses to the non-sensitive control items is first modeled as a function of covariates, and the resulting fitted values are then used in a second stage to estimate the probability of endorsing the sensitive item, allowing covariate effects while maintaining respondent privacy under standard identifying assumptions. The Tsai (2019) Stata implementation operationalizes this approach by providing least-squares and maximum-likelihood estimators that follow the same two-stage logic and automate inference.

list experiment. Two out of the three estimates are positive, but they are not precise and not statistically significant.

4.2.4 Effects at the District Level Measured with Standard Difference-in-Differences

Our main empirical specification is an age-cohort difference-in-differences, where we study how cohorts' exposure to university openings affects these cohorts' experiences and perceptions of corruption. In Appendix Figures [A.6](#), [A.7](#), and [A.8](#), we instead use a standard difference-in-differences specification, utilizing variations created by districts with university openings that happened during the period of our data collection.²⁷ We do not find evidence that university openings affect corruption exposure, perceptions, or bribes paid for specific procedures or services, with combined event study coefficients reported in Appendix Tables [A.6](#), [A.7](#), and [A.8](#). While these results allow us to answer the policy-relevant question of “what happens to corruption after a university is opened?” they must be interpreted carefully. It also takes much longer for a university to increase the levels of education in the general population than it does for a specific cohort. Therefore, the results cannot be interpreted as showing the effect of educating a whole population on corruption. Nonetheless, the results are consistent with our main results that education does not decrease, and may instead increase, corruption at the individual- or cohort-level, and we find no evidence that opening a university reduces corruption at the district-level in the short- or medium-run.

²⁷The identifying assumption for this analysis is that districts with and without university openings are on parallel trends in terms of our corruption measures.

4.3 Robustness Tests

We show that our main result is robust to a variety of specifications. First, our stacked difference-in-differences model, detailed in Section 3.2, accounts for potential treatment effect heterogeneity when university opening timing is staggered across districts. We create sub-datasets with only districts in G_g (treatment group with university opening year g) and the never-treated districts (control group), similar to Cengiz et al. (2019), which enables a clean comparison between the treatment group G and a never-treated comparison group in each year.²⁸ Appendix Figure A.9 and Appendix Figure A.10 show that results from the stacked regressions are similar to our baseline results.

We report the parallel pre-trends test using Rambachan and Roth (2023)’s approach in Appendix Figure A.11. We use stacked regression described in Equation 3 to test for the effect in the first post-period in panels (a), (c), and (e), and the average causal effects in panels (b), (d), and (f). The test of the average causal effect across all seven post-periods in panels (b), (d), and (f) is more stringent than the default test for the first period because the treatment and control groups have more time to diverge.²⁹ Our estimates for the effect of the university expansion on education are robust to differential trends that are equal to the largest differential pre-treatment trends ($\bar{M} = 1$) in the first post-treatment period (panel (a)) as well as in all post-treatment periods (panel (b)). Although the robust confidence intervals for our corruption measures exclude 0 only with a less stringent restriction (panels (c) and (e)), the more stringent tests in panels (d) and (f) also do not suggest a negative effect on corruption exposure.

²⁸We also contrast the stacked difference-in-differences estimator with other conventional staggered difference-in-differences estimators and explain our choice in Section 3.2.

²⁹The identified set for the second period will be twice as large as for the first period, three times as large in the third period, and so on (Rambachan and Roth, 2023).

Moreover, we report the absence of pre-trends and robust results of our main education and bribe request outcomes for different regions across Vietnam in Appendix Figure A.12, where we restrict the sample to the North, Middle, or South region. In Appendix Figure A.13, we show that additional controls of provincial trends yield results similar to our baseline results.

As both the university expansion and our outcome variables could be spatially correlated, we show the robustness of our main education and corruption exposure results with Conley-adjusted standard errors in Appendix Tables A.9 and A.10.

Throughout the paper, we use the 23 to 27 age cohort as the reference group for individual outcomes and the 28 to 32 age cohort as the reference group for household-level outcomes. In Appendix Figures A.14 and A.15, we show that using the 28 to 32 age cohort as the reference group for the individual outcomes does not change our main results. Finally, we assess the robustness of our estimates to different sizes of the age bin. Appendix Figure A.16 additionally presents the corruption exposure results using age bins of three and age bins of two. The main patterns are very similar.

5 Discussion and Potential Mechanisms

Overall, our results show a dramatic increase in university education as a consequence of opening new universities. However, this increase in education did not translate into a decrease in corruption. Instead, we observe substantial increases in both self-reported incidences of corruption and perceptions of corruption. We now turn to potential explanations for the lack of a decrease in corruption and mechanisms for the increase in corruption exposure.

These findings point to two potentially countervailing mechanisms connecting education to corruption at the individual level. The first is the *knowledge effect*, which has

received the most attention in the literature ([Lipset, 1960](#); [Svensson, 2005](#); [Glaeser and Saks, 2006](#)). According to this logic, education equips citizens with greater knowledge and capabilities to avoid corruption. Furthermore, it is more risky for government officials to ask for bribes from educated individuals as they have better means to report the government officials to the authorities. For this reason, educated actors should also be less likely to be targeted by officials for bribe requests ([Fried et al., 2010](#)).

However, our findings above also suggest an *income effect* that may push against and even overwhelm the knowledge effect ([Mocan, 2008](#)). Higher incomes, generated by increased education, generate both greater exposure to situations in which bribes may be requested and may also raise the relative value of paying these bribes. Those with higher incomes are more likely to engage in activities, such as buying land, expanding a residence, or obtaining a driver's license, that expose them to more bribe requests from regulators than those with lower incomes. Higher incomes may also raise the relative value of an individual's time, as time away from the job is more costly. Therefore, burdensome administrative procedures and long waiting periods become more consequential, potentially making wealthier individuals more willing to pay bribes to avoid these costs. If this higher willingness-to-pay is observable for those requesting bribes, higher incomes may also increase bribe requests as part of systematic price (or bribe) discrimination ([Olken and Barron, 2009](#); [Mattsson, 2025a](#)).

Therefore, the direction of the effect of education on bribery depends on the setting. The knowledge effect is likely to be dominant where accountability channels are strong, but education is required to access these channels. For example, if there is effective anti-corruption legislation and institutions where citizens can report corrupt

officials.³⁰

On the other hand, the income effect is likely to dominate in settings without effective accountability channels or where such channels are thought to be illegitimate and distrusted. Knowledge about denunciation and complaint mechanisms is of little value when citizens believe that the government has limited political will to punish corrupt actors (Brinkerhoff, 2000; Ankamah and Manzoor E Khoda, 2018).

In this section, we attempt to demonstrate the role of these two mechanisms. First, we test whether the knowledge effect exists in our setting by establishing whether university education increases knowledge about the policy prescriptions in the Anti-Corruption Law. Second, we test if more education also increases the likelihood of individuals denouncing corrupt officials. Third, we trace the income effect by establishing that university opening had a positive effect on income and that higher-income individuals were more likely to bribe and, to save time, hire middlemen to facilitate their bribe payments. Fourth, we take advantage of the 2016 Blazing Furnace anti-corruption campaign to see if the highly publicized punishments of corrupt officials and promotion of mechanisms for disputing bribes, along with e-governance channels for avoiding bureaucratic discretion, altered the relative importance of the knowledge and income effects in Vietnam. We finish the section by ruling out several reasonable alternative mechanisms.

5.1 The Knowledge Effect

We will now test the hypothesis that a more educated population would make more complaints against corrupt officials and thereby hold them accountable (Svensson,

³⁰The institutions could include an ombudsman and complaint mechanisms for bureaucratic corruption, courts to sue corrupt officials, and government inspectorates who can investigate and punish corrupt acts. In these cases, educated citizens will have greater knowledge that the institutions exist and a greater capacity to avail themselves of them (Grossman and Slough, 2022).

2005; Botero et al., 2013). Appendix Table A.11 shows suggestive evidence that education raises knowledge about Vietnam’s anti-corruption legislation, although the levels of knowledge are generally low and the estimated effects are imprecise. Figure 5 shows the effect of the university expansion on citizens denouncing government officials asking for bribes. Panel (a) of Figure 5 shows the effect on the probability of denouncing a government official after a hypothetical bribe request, which every respondent answers. Panel (b) shows the effect on the likelihood of actually denouncing a government official, among households that were asked for a bribe. Panel (c) shows the same analysis as panel (b) using the whole sample. Panel (c) hence takes into account both the effect on the probability of having been asked for a bribe and the effect on the probability of denouncing the government official after having been asked to pay a bribe. Appendix Tables A.12 and A.13 report coefficient estimates. Overall, we don’t find any effects of the university openings on the reporting of government officials for corrupt behaviors for any of the three measures. This is evidence against the hypothesis that education makes people more likely to report corrupt officials, one of the mechanisms through which education has been hypothesized to reduce corruption at the societal level.

5.2 The Income Effect

In Figure 6 and Appendix Table A.14, we demonstrate a significant 17.4% increase in households’ monthly income for cohorts exposed to the university opening. Additionally, in Appendix Figure A.17, we find a positive association between household income and corruption exposure. Although the relationship between income and bribery should not be interpreted as causal, this suggests that more educated families may bribe more as they become richer.

One reason why individuals with a higher income may engage in more corruption is that they face higher opportunity costs for their time. They may therefore be more willing to pay to avoid the administrative hassle. We test this hypothesis by focusing on payments to “middlemen” who help out with applications for government procedures. While paying a middleman does not necessarily involve paying a bribe, in practice, it is common for the middleman to pay the government official. Appendix Figure A.18 and Appendix Table A.15 present suggestive evidence that university education leads to an increase in payments to middlemen. This is consistent with some of the increase in corruption being driven by increases in the opportunity cost of time for those with a university education.

Another way in which higher incomes may lead to bribery is by increasing the number of interactions the individual has with potentially corrupt government officials. For example, individuals may be more likely to apply for construction permits as income increases. In Appendix Figure A.19 and Appendix Table A.16, we analyze the effect of education on the number of government procedures (certifications, construction permits, land use rights, and other administrative procedures) and services (public healthcare and primary school) that the individual applied for or used. For this group of government interactions, we rule out that individuals have more interactions with the government officials following the university expansion. Instead, we see a decrease in the use of public services, which is not surprising given that the increase in income is likely to have caused a switch from public to private providers of these services. While the survey data does not cover all of the potential interactions with government officials, it does cover some of the most common types of interactions where individuals often pay bribes. However, it is possible that education, and subsequently income, increased interactions with government officials that are not covered by the survey, and that an increase in these interactions is a contributing factor to

the increase in experienced and perceived corruption.

In Appendix Figures [A.6](#), [A.7](#), and [A.8](#), we use a standard difference-in-differences specification, utilizing variations created by districts with university expansion that happened during the survey waves, and we do not find an increase in corruption exposure or perceptions for the entire district population.³¹ These results are consistent with a targeting of higher education (and therefore high-income) individuals, rather than an increase in corruption across all of society when a new university is established.

5.3 Did the Anti-corruption Campaign Increase the Importance of the Knowledge Effect?

In Appendix Table [A.17](#), we analyze the differences in the effects of education before and after the start of the Anti-Corruption Campaign described in Section [2.1](#) and Appendix [A.1](#). In Column (1), we show that the first-stage effect of university expansion on corruption is very similar before and after the anti-corruption campaign. In Column (2), we show that the effect of education on knowledge about the Law on Corruption Prevention was larger after the start of the anti-corruption campaign. The difference in effects is only marginally statistically significant, but it is intuitive that more educated individuals gather more knowledge about the government’s anti-corruption efforts once they believe that these efforts are sincere. In Column (3), we show that the effect of education on actually denouncing corrupt behaviors is more positive after the anti-corruption campaign started. In columns (4) and (5), we show that the effects of education on corruption are not statistically different after the start

³¹These results should be interpreted cautiously as described in detail in Section [4.2.4](#).

of the corruption campaign.³² Taken together, these results tentatively indicate that the relative magnitudes of the knowledge effect and income effect depend on institutions and the trust the population has in the government’s commitment to combating corruption. When the institutions are more committed to combating corruption, the knowledge effect starts to counteract the income effect, as educated individuals are more likely to denounce corrupt behaviors. However, since the differences in effects on corruption experiences are not statistically significant, they should be interpreted very cautiously.

5.4 Alternative Mechanisms

We also test for two other potential mechanisms. First, it is possible that new universities led to better-educated government officials and that the officials with higher education are more corrupt. Appendix Figure A.20 restricts the PAPI sample to those who work as government officials and measures the effect of exposure to a university opening on the education level of government officials. We do not find any evidence of an increase in education levels among government officials, and thus, it is unlikely that the education levels of government officials are driving the effect on corruption.

Second, we also rule out the potential mechanisms that migration into districts with a university is driving our effects. Appendix Figure A.21 shows the effect of exposure to a university opening on the share of migrants in our sample. We do not find any evidence that exposure to a university opening is associated with being a

³²Appendix Table A.18 shows the results before and after the start of the anti-corruption campaign for our other corruption outcomes. We see an insignificant positive effect on corruption perceptions after the start of the corruption campaign. This may be explained by respondents with higher education being more aware of the corruption cases revealed in the campaign. For corruption during specific interactions with government officials, we do not find any evidence of a difference in the effect before and after the start of the anti-corruption campaign, but the estimates are relatively imprecise.

migrant.³³ Vu (2025) also found that the higher education expansion in Vietnam has little effect on both in-migration and out-migration using the Labor Force Survey, which is more suitable for studying migration effects than the PAPI data.

6 Conclusion

In this paper, we explore the impact of increased university education on corruption, leveraging Vietnam’s university expansion, detailed survey data and an age-cohort difference-in-differences approach. Despite a significant increase in university attendance among the affected cohorts, our findings indicate that this educational uplift does not lead to a reduction in corruption for the more educated cohorts. In fact, the share affected by corruption increased among individuals exposed to the university expansion.

Our analysis revealed that the higher likelihood of bribery among the educated cohorts is associated with higher household incomes, which correlate with a greater propensity to pay bribes. This suggests that while education raises income levels, it also inadvertently increases opportunities and capacities for corrupt transactions, contradicting the assumption that education inherently reduces corruption.

Our paper has several limitations. First, we are mostly limited to studying the effects of increasing the education levels of individual cohorts on the corruption levels that these cohorts are exposed to. It is possible these effects are different from the effects of education on corruption at a societal level, where the whole population is more educated. We expand the analysis to measure district-level effects and these

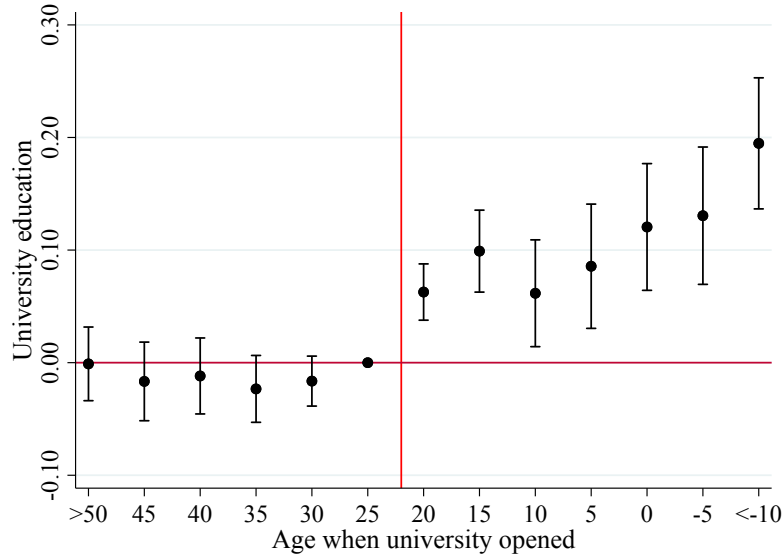
³³We do not claim that there is no effect of university openings on migration because the way the PAPI survey respondents are sampled is not suitable for such an analysis. The PAPI sample is mostly drawn from the population of permanent residents. However, even though we are unable to measure the effect on migration overall, we can still rule out the potential mechanism of new migrants driving the effects we see in our PAPI sample.

results are also consistent with there being no negative effect of education on corruption, but as most of our variation comes from the differences between cohorts, these results are less precisely measured. Furthermore, we investigate some of the proposed mechanisms for why education may reduce corruption at a societal level, such as increasing the rate at which citizens report corrupt government officials. Second, our measures of corruption are self-reported, and there is always a risk that they are biased due to social desirability or fear of being held accountable for giving bribes. It is reassuring that results are consistent with using a list experiment to reduce such bias, but the list experiment questions are only available for a small share of our outcome variables. Third, we can only directly measure the effect on small-scale corruption between citizens and government officials. For larger-scale corruption taking place at higher levels of government, we only have data on citizens' perceptions. Furthermore, for education to shift larger-scale corruption activities, it would probably be necessary to increase education at the national-level, as opposed to at the individual, cohort, or local level.

Finally, our setup allows us to examine both the short-run and long-run effects of university expansion on exposure to, perception of, and engagement in corruption. However, as is indicated in our discussion section, it is possible that these results are contingent on the specific institutional context. Specifically, we show suggestive evidence that as the institutional context changed to one where the government took a stronger anti-corruption position, the knowledge effect causes those with more education to be more likely to report corrupt government officials. While our findings cast doubt on whether increasing access to education is an effective tool for combating corruption by fostering more educated younger cohorts, they also point to a complex relationship between education, income, and corrupt behavior.

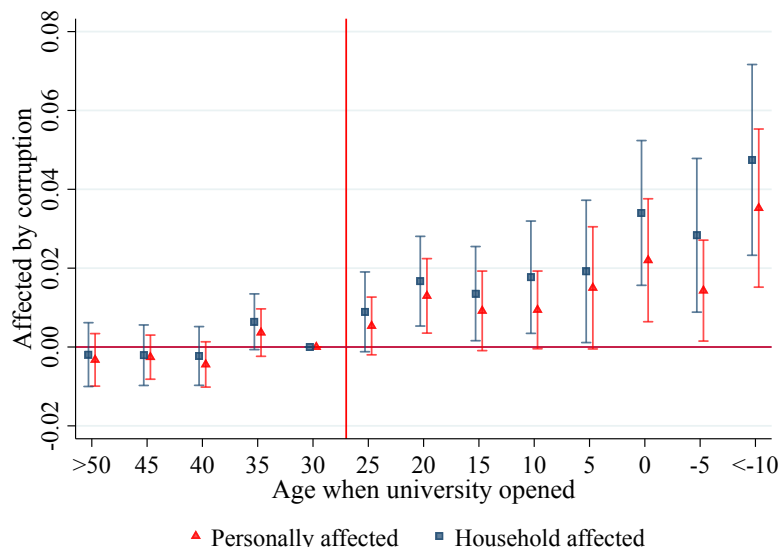
Figures

Figure 1: Effect of university expansion on respondents receiving university education



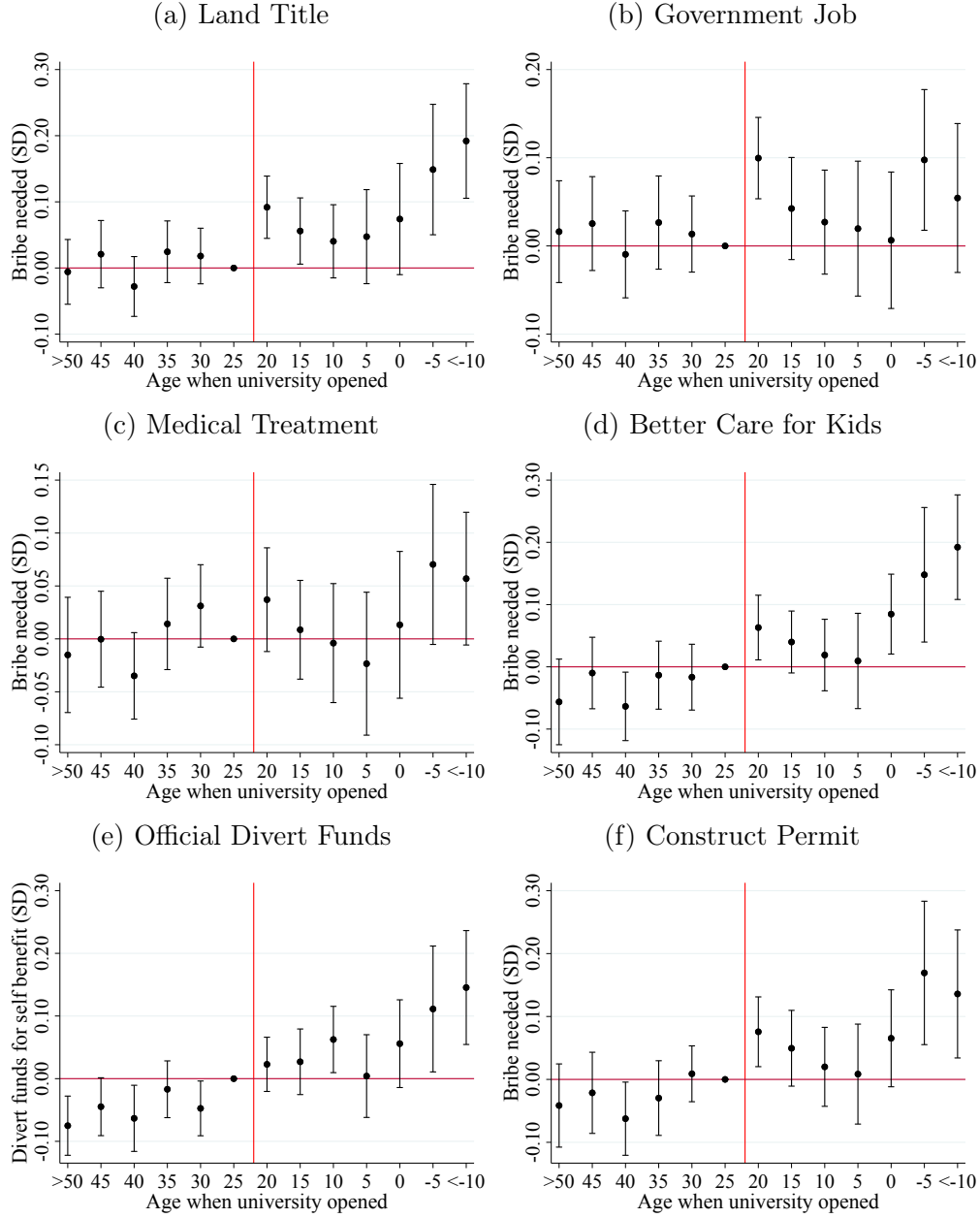
Note: Figure 1 shows the effect of cohorts being exposed to a university opening on the share receiving some university education. We use Equation 1 to obtain the estimates for each cohort. The outcome variable is an indicator of the respondent being university-educated. As the outcome is at the individual level, the reference age group is between 23 to 27, which is explained in Section 3. The mean of the outcome variable in districts that never had a university is 0.112. The number of observations is 171,162. 95% confidence intervals are constructed using standard errors clustered at the district level. The difference-in-differences estimate for all cohorts is reported in Table 1. Appendix Figure A.3 reports the effect on university completion. The stacked difference-in-differences estimates, which account for the negative weight issue as explained in Section 3.2, are reported in Appendix Figure A.9. See discussion in Section 4.1.

Figure 2: Effect of university expansion on whether the household was affected by bribe-taking



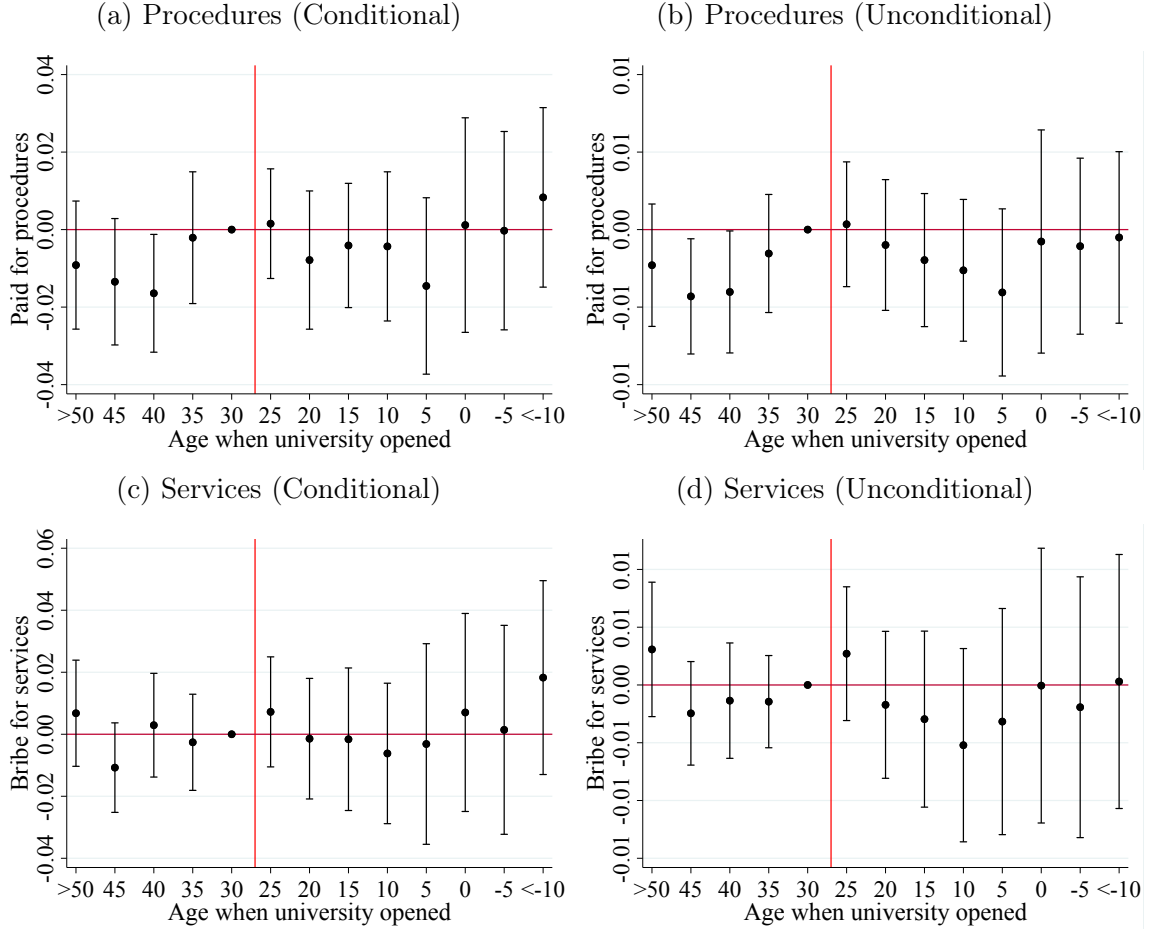
Note: Figure 2 shows the effect of cohorts being exposed to a university opening on the share having experienced an act of corruption in the previous year. We use Equation 1 to obtain the estimates for each cohort ($N=169,456$). The outcome variable “personally affected” is an indicator of being personally affected by an act of bribe-taking, and “household affected” is an indicator of being personally affected or having a family member affected. The survey question is “Have you or anyone in your household been affected by an act of bribe-taking by a government official in the past year? Consider all types of bribe-giving, ranging from giving a small bribe to a traffic official to giving a commission in order to get a government contract.” This outcome is at the household level, as younger family members of those aged between 23 to 27 might be exposed to university expansion, which affects the household through spillover effects. Therefore, as explained in Section 3, the reference age group is between 28 to 32. Appendix Figure A.4 shows a consistent result using the age cohort of 23 to 27 as the reference group. The means of the outcome variables in districts that never had a university are 0.018 and 0.029 for personal and household experiences, respectively. 95% confidence intervals are constructed using standard errors clustered at the district level. The difference-in-differences point estimate for all cohorts is reported in Table 2. The stacked difference-in-differences estimates, which account for the negative weight issue as explained in Section 3.2, are reported in Appendix Figure A.10. See discussion in Section 4.2.

Figure 3: Effect of university expansion on respondents' perceptions of corruption



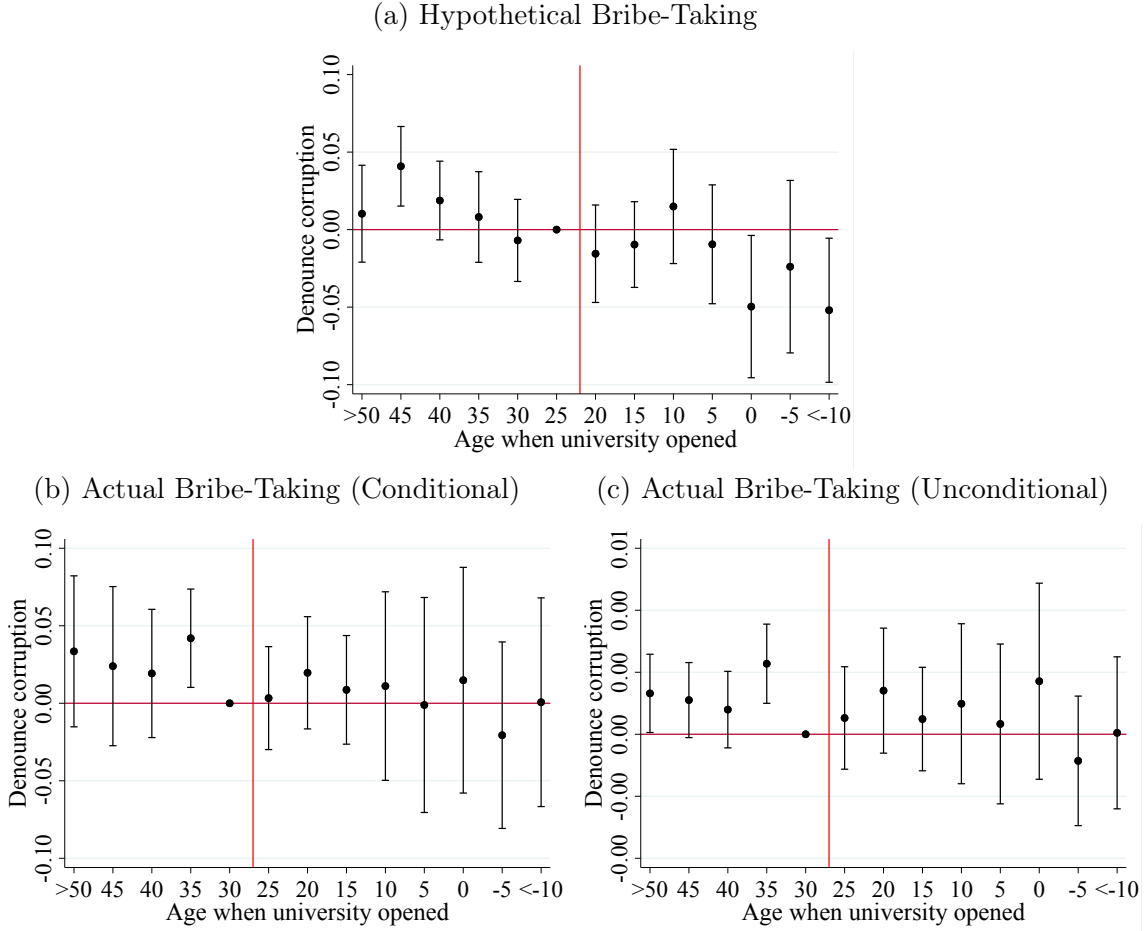
Note: Figure 3 shows the effect of cohorts being exposed to a university opening on the respondents' perception of corruption. We use the specification in Equation 1 to obtain estimates for each cohort. The outcome variables are standardized measures of respondents' agreement with different statements related to corruption perceptions (see Appendix A.5 for details). The respondents are asked about their own perceptions, so the result is at the individual level, with the reference age group between 23 to 27. For panels (a), (b), (c), (d), (e), and (f), number of observations is 140,200, 138,407, 149,958, 149,588, 135,848, 138,593. Control means are -0.006, -0.032, -0.009, -0.046, 0.009, -0.053. 95% confidence intervals are constructed using standard errors clustered at the district level. The difference-in-differences estimates are reported in Table 3. See discussion in Section 4.2.

Figure 4: Effect of university expansion on informal payments during interactions with government



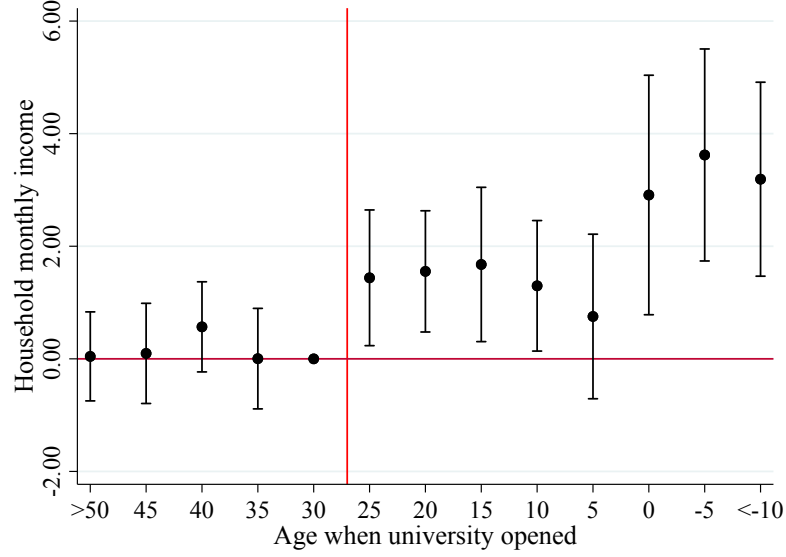
Note: Figure 4 shows the effect of cohorts being exposed to a university opening on households making informal payments. We use Equation 1 to obtain estimates for each cohort. The outcome variable is an indicator variable if an individual or an individual's family paid a bribe or a middleman as part of an interaction with the government. As the outcome is at the household level, the reference age group is between 28 to 32, which is explained in Section 3. The government procedures in Panels (a) and (b) are applications for a certification, construction permit, land use rights, or other administrative procedures. The government services in panels (c) and (d) are public primary schools or public hospitals. Panels (a) and (c) are conditional on having had the interaction ($N=100,434$ and $N=85,149$, respectively). Panels (b) and (d) use all observations ($N=171,162$). The means of the outcome variable in districts that never have a university are 0.105, 0.023, 0.092, and 0.028 respectively for panels (a), (b), (c), and (d). 95% confidence intervals are constructed using standard errors clustered at the district level. The difference-in-differences estimates are reported in Appendix Table A.3. See discussion in Section 4.2.

Figure 5: Effect of university expansion on respondents denouncing corruption



Note: Figure 5 shows the effect of cohorts being exposed to a university opening on denouncing acts of corruption. We use Equation 1 to obtain the event study estimates. The outcome variable “denounce corruption” is an indicator for attempting to denounce the act of corruption. Panel (a) uses the survey question “If you found yourself the victim of a bribe request, [w]ould you be willing to denounce the bribe if it was [amount]?” answered by respondents from 2015 to 2021 ($N=93,720$). The program randomly chooses one amount, and thus, we additionally control for the bribe amount fixed effect for this outcome. As the outcome is at the individual level, the reference age cohort is 23 to 27. The PAPI survey question for panels (b) and (c) is “Did you or your family member attempt to denounce this act of corruption?” Panel (b) performs the analysis on a subset of observations ($N=5,517$) conditional on being affected by bribe-taking acts, while panel (c) is unconditional on whether being affected or not ($N=171,162$). These outcomes are at the household level based on a follow-up question for households being affected by bribery, an outcome used in Figure 2. Therefore, the reference age cohort is 28 to 32 for panels (b) and (c). The mean of the indicator in the panel for districts that never have a university is 0.598 in panel (a), 0.069 in panel (b), and 0.002 in panel (c). 95% confidence intervals are constructed using standard errors clustered at the district level. The combined difference-in-differences point estimates for the younger cohorts are reported in Appendix Table A.12 for panel (a) and Appendix Table A.13 for panels (b) and (c). See discussion in Section 5.

Figure 6: Effect of university expansion on household income



Note: Figure 6 shows the effect of cohorts being exposed to a university opening on household income. We use Equation 1 to obtain the event study estimates. The outcome variable “Household monthly income” is in million VND. As the outcome is at the household level, the reference age cohort is 28 to 32. The number of observations is 44,529. The mean of the household income for districts that never have a university is 7.535 million VND. 95% confidence intervals are constructed using standard errors clustered at the district level. The combined difference-in-differences point estimates for the younger cohorts are reported in Appendix Table A.14. See discussion in Section 5.

Tables

Table 1: The effect of university expansion on respondents receiving some university education

	University Education		
	(1)	(2)	(3)
University Expansion	0.088*** (0.012)	0.085*** (0.012)	0.094*** (0.013)
Observations	171,162	171,162	171,162
Cluster No.	320	320	320
Adjusted R2	0.110	0.116	0.314
Mean of Never Treated	0.112	0.112	0.112
District FE	✓	✓	✓
Year FE	✓	✓	✓
Age FE	✓	✓	✓
Demographic Controls		✓	✓
Full Controls			✓

Note: Table 1 shows the effect of the university expansion on the share of individuals with some university education. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 22 or below at the time when the university opened and 0 otherwise. The individual-level outcome variable is the same as that in Figure 1. “Demographic controls” include gender and ethnicity. “Full controls” additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. For Conley-adjusted standard errors, see Appendix Table A.9. See discussion in Section 4.1.

Table 2: The effect of university expansion on whether the household was affected by bribe-taking

	Personally Affected			Household Affected		
	(1)	(2)	(3)	(4)	(5)	(6)
University Expansion	0.0093** (0.0037)	0.0090** (0.0037)	0.0089** (0.0037)	0.0121*** (0.0046)	0.0117** (0.0046)	0.0117** (0.0046)
Observations	169,456	169,456	169,456	169,456	169,456	169,456
Cluster No.	320	320	320	320	320	320
Adjusted R2	0.016	0.018	0.018	0.020	0.022	0.022
Never T. Mean	0.018	0.018	0.018	0.029	0.029	0.029
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Age FE	✓	✓	✓	✓	✓	✓
Demo. Controls		✓	✓		✓	✓
Full Controls			✓			✓

Note: Table 2 shows the effect of exposure to the university expansion on the household’s probability of experiencing an act of bribe-taking. The outcome variable “personally affected” is an indicator of being personally affected by an act of bribe-taking, and “household affected” is an indicator of being personally affected or having a family member affected. The outcome variable is the same as that in Figure 2. This outcome is at the household level, as younger family members of those aged between 23 to 27 are exposed to university expansion, which affects the household through spillover effects. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 27 or below at the time when the university opened and 0 otherwise. “Demographic controls” include gender and ethnicity. “Full controls” additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. For Conley-adjusted standard errors, see Appendix Table A.10. See discussion in Section 4.2.

Table 3: The effect of university expansion on respondents' corruption perceptions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Land	Govt Job	Medical	Kids	Funds	Permit	Index
University Expansion	0.062*** (0.019)	0.051** (0.020)	0.013 (0.018)	0.065*** (0.021)	0.062*** (0.017)	0.072*** (0.021)	0.057*** (0.019)
Observations	140,200	138,407	149,958	149,588	135,848	138,593	163,744
Cluster No.	320	320	320	320	320	320	320
Adjusted R2	0.038	0.114	0.075	0.041	0.018	0.051	0.061
Mean of Never Treated	-0.006	-0.032	-0.009	-0.046	0.009	-0.053	-0.026
District FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Age FE	✓	✓	✓	✓	✓	✓	✓

Note: Table 3 shows the effect of university expansion on respondents' perception of several corrupt behaviors, such as "people have to bribe to obtain a land title", or "for a government job", "to receive medical treatment in the public hospitals", or "have their children better attended", or "officials divert funds for their personal benefit", or to "obtain a construction permit". Respondents choose an option among "agree", "somewhat agree", and "disagree". We standardize the outcome variables to have a mean of zero and a standard deviation of one. The overall index in column 7 is a standardized weighted average of these standardized components, which also increases as a result of university expansion (see Appendix A.5 for details of the outcome construction). These outcome variables are the same as those in Figure 3. Since the respondents are asked about their own perceptions, these are individual-level outcomes. We combine the age cohort dummies in Equation 1 into an independent variable "University Expansion" which equals 1 if the survey participant was aged 22 or below at the time when the university opened and 0 otherwise. Standard errors are clustered at the district level. We test the robustness of the corruption perception index in Appendix Table A.2 with additional controls. See discussion in Section 4.2.

References

- D. Acemoglu, S. Johnson, J. A. Robinson, and P. Yared. From education to democracy? *American Economic Review*, 95(2):44–49, 2005.
- D. Acemoglu, S. Johnson, J. A. Robinson, and P. Yared. Reevaluating the modernization hypothesis. *Journal of monetary economics*, 56(8):1043–1058, 2009.
- D. Acemoglu, S. Naidu, P. Restrepo, and J. A. Robinson. Democracy does cause growth. *Journal of political economy*, 127(1):47–100, 2019.
- M. Agerberg. Corrupted estimates? Response bias in citizen surveys on corruption. *Political Behavior*, 44(2):653–678, 2022.
- M. L. Anderson. Multiple inference and gender differences in the effects of early intervention: A reevaluation of the abecedarian, perry preschool, and early training projects. *Journal of the American Statistical Association*, 103(484):1481–1495, 2008.
- J. C. Andvig and K. O. Moene. How corruption may corrupt. *Journal of Economic Behavior & Organization*, 13(1):63–76, 1990.
- S. S. Ankamah and S. Manzoor E Khoda. Political will and government anti-corruption efforts: What does the evidence say? *Public Administration and Development*, 38(1):3–14, 2018.
- M. N. Asadullah, L. D. H. Perera, and S. Xiao. Vietnam’s extraordinary performance in the PISA assessment: A cultural explanation of an education paradox. *Journal of Policy Modeling*, 42(5):913–932, 2020.
- J. Bai, S. Jayachandran, E. J. Malesky, and B. A. Olken. Firm growth and corruption: Empirical evidence from Vietnam. *The Economic Journal*, 129(618):651–677, 2019.
- A. Banerjee, E. Duflo, C. Imbert, S. Mathew, and R. Pande. E-governance, accountability, and leakage in public programs: Experimental evidence from a financial management reform in India. *American Economic Journal: Applied Economics*, 12(4):39–72, 2020.
- G. Blair and K. Imai. Statistical analysis of list experiments. *Political Analysis*, 20(1):47–77, 2012.
- J. Botero, A. Ponce, and A. Shleifer. Education, complaints, and accountability. *The Journal of Law and Economics*, 56(4):959–996, 2013.
- D. W. Brinkerhoff. Assessing political will for anti-corruption efforts: An analytic framework. *Public Administration and Development: The International Journal of Management Research and Practice*, 20(3):239–252, 2000.
- B. Callaway and P. H. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.

- F. R. Campante and D. Chor. Schooling, political participation, and the economy. *Review of Economics and Statistics*, 94(4):841–859, 2012.
- F. R. Campante and Q.-A. Do. Isolated capital cities, accountability, and corruption: Evidence from US states. *American Economic Review*, 104(8):2456–2481, 2014.
- D. Cengiz, A. Dube, A. Lindner, and B. Zipperer. The effect of minimum wages on low-wage jobs. *The Quarterly Journal of Economics*, 134(3):1405–1454, 2019.
- Centre for Community Support and Development Studies, Centre for Research and Training of the Viet Nam Fatherland Front (VFF-CRT), Real-Time Analytics (RTA), and United Nations Development Programme (UNDP). The 2022 Viet Nam provincial governance and public administration performance index (PAPI 2022): Measuring citizens’ experiences. Final report, 2023.
- K. Croke, G. Grossman, H. A. Larreguy, and J. Marshall. Deliberate disengagement: How education can decrease political participation in electoral authoritarian regimes. *American Political Science Review*, 110(3):579–600, 2016.
- H.-A. Dang, P. Glewwe, J. Lee, and K. Vu. What explains Vietnam’s exceptional performance in education relative to other countries? Analysis of the 2012, 2015, and 2018 PISA data. *Economics of Education Review*, 96:102434, 2023.
- C. De Chaisemartin and X. d’Haultfoeuille. Two-way fixed effects estimators with heterogeneous treatment effects. *American economic review*, 110(9):2964–2996, 2020.
- T. S. Dee. Are there civic returns to education? *Journal of public economics*, 88(9-10):1697–1720, 2004.
- Q.-A. Do, T. Van Nguyen, and A. Tran. Do people pay higher bribes for urgent services? Working paper, 2013.
- C. Ferraz, F. Finan, and D. B. Moreira. Corrupting learning: Evidence from missing federal education funds in Brazil. *Journal of Public Economics*, 96(9-10):712–726, 2012.
- R. Fisman and M. A. Golden. *Corruption: What everyone needs to know*. Oxford University Press, 2017.
- B. J. Fried, P. Lagunes, and A. Venkataramani. Corruption and inequality at the crossroad: A multimethod study of bribery and discrimination in Latin America. *Latin American Research Review*, 45(1):76–97, 2010.
- N. K. Giang. The political economy of Vietnam’s anti-corruption campaign. *Southeast Asian Affairs*, pages 375–390, 2024.
- E. L. Glaeser and R. E. Saks. Corruption in America. *Journal of Public Economics*, 90(6-7):1053–1072, 2006.

- E. L. Glaeser, R. La Porta, F. Lopez-de Silanes, and A. Shleifer. Do institutions cause growth? *Journal of Economic Growth*, 9:271–303, 2004.
- E. L. Glaeser, G. A. Ponzetto, and A. Shleifer. Why does democracy need education? *Journal of economic growth*, 12(2):77–99, 2007.
- A. Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021.
- G. Grossman and T. Slough. Government responsiveness in developing countries. *Annual Review of Political Science*, 25(1):131–153, 2022.
- E. A. Hanushek and L. Woessmann. The knowledge capital of nations: Education and the economics of growth. 2015.
- L. Hendricks and T. Schoellman. Skilled labor productivity and cross-country income differences. *American Economic Journal: Macroeconomics*, 15(1):240–268, 2023.
- N. M. Hung. Vietnam in 2017: Power consolidation, domestic reforms, and coping with new geopolitical challenges. *Southeast Asian Affairs*, pages 407–428, 2018.
- H. Larreguy and J. Marshall. The effect of education on civic and political engagement in nonconsolidated democracies: Evidence from nigeria. *Review of Economics and Statistics*, 99(3):387–401, 2017.
- S. M. Lipset. *Political Man: The Social Bases of Politics*. Doubleday Company, 1960.
- R. E. Lucas Jr. On the mechanics of economic development. *Journal of Monetary Economics*, 22(1):3–42, 1988.
- E. Malesky, P. Ngoc, C. Chen, and L. W. Meredith. Rust removal: Why Vietnam’s anti-corruption efforts have failed to deliver results and what that implies for future campaigns. *The Political Logics of Anticorruption Efforts in Asia*, pages 129–73, 2019.
- E. Malesky, N. T. Pham, and T. N. Phan. The Vietnam provincial competitiveness index and provincial green index report: Promoting a business-enabling and environmentally friendly investment climate, 2023. Final report, Vietnam Chamber of Commerce and Industry and United States Agency for International Development, 2024.
- E. J. Malesky and T. H. Bui. Vietnam in 2023: Stoking the embers of the “blazing furnace”. *Southeast Asian Affairs*, 2024.
- E. J. Malesky, D. D. Gueorguiev, and N. M. Jensen. Monopoly money: Foreign investment and bribery in Vietnam, a survey experiment. *American Journal of Political Science*, 59(2):419–439, 2015.
- M. Mattsson. When does corruption cause red tape? Bribe discrimination under

- asymmetric information. *Journal of Public Economics*, 250:105483, 2025a.
- M. Mattsson. Information systems, service delivery, and corruption: Evidence from the Bangladesh civil service. *American Economic Journal: Applied Economics*, 17(3):414–445, 2025b.
- S. McGuinness, E. Kelly, T. T. P. Pham, T. T. T. Ha, and A. Whelan. Returns to education in Vietnam: A changing landscape. *World Development*, 138:105205, 2021.
- K. Milligan, E. Moretti, and P. Oreopoulos. Does education improve citizenship? evidence from the United States and the United Kingdom. *Journal of public Economics*, 88(9-10):1667–1695, 2004.
- N. Mocan. What determines corruption? International evidence from microdata. *Economic Inquiry*, 46(4):493–510, 2008.
- B. A. Olken. Monitoring corruption: Evidence from a field experiment in Indonesia. *Journal of Political Economy*, 115(2):200–249, 2007.
- B. A. Olken. Corruption perceptions vs. corruption reality. *Journal of Public Economics*, 93(7-8):950–964, 2009.
- B. A. Olken and P. Barron. The simple economics of extortion: Evidence from trucking in Aceh. *Journal of Political Economy*, 117(3):417–452, 2009.
- B. A. Olken and R. Pande. Corruption in developing countries. *Annu. Rev. Econ.*, 4(1):479–509, 2012.
- A. Rambachan and J. Roth. A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591, 2023.
- R. Reinikka and J. Svensson. Fighting corruption to improve schooling: Evidence from a newspaper campaign in Uganda. *Journal of the European economic association*, 3(2-3):259–267, 2005.
- R. Reinikka and J. Svensson. The power of information in public services: Evidence from education in uganda. *Journal of Public Economics*, 95(7-8):956–966, 2011.
- P. M. Romer. Endogenous technological change. *Journal of political Economy*, 98(5, Part 2):S71–S102, 1990.
- J. Roth, P. H. Sant’Anna, A. Bilinski, and J. Poe. What’s trending in difference-in-differences? a synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2):2218–2244, 2023.
- S. Ruggles, L. Cleveland, R. Lovaton, S. Sarkar, M. Sobek, D. Burk, D. Ehrlich, Q. Heimann, J. Lee, and N. Merrill. Integrated public use microdata series, international: Version 7.6, 2025. URL <https://doi.org/10.18128/D020.V7.6>.
- S. Sharma, S. Singhal, and F. Tarp. Corruption and mental health: Evidence from

- Vietnam. *Journal of Economic Behavior & Organization*, 185:125–137, 2021.
- R. M. Sondheimer and D. P. Green. Using experiments to estimate the effects of education on voter turnout. *American Journal of Political Science*, 54(1):174–189, 2010.
- J. Svensson. Eight questions about corruption. *Journal of Economic Perspectives*, 19(3):19–42, 2005.
- D. Treisman. The causes of corruption: A cross-national study. *Journal of Public Economics*, 76(3):399–457, 2000.
- D. Treisman. What have we learned about the causes of corruption from ten years of cross-national empirical research? *Annu. Rev. Polit. Sci.*, 10(1):211–244, 2007.
- C.-l. Tsai. Statistical analysis of the item-count technique using Stata. *The Stata Journal*, 19(2):390–434, 2019.
- A. Valero. Education and economic growth. In *The Routledge Handbook of the Economics of Education*, pages 555–582. Routledge, 2021.
- H. L. Vu and T. A. Nguyen. Analysis of access and equity in higher education system in Vietnam. *VNU Journal of Science: Policy and Management Studies*, 34(4), 2018.
- K. Vu. Higher education expansion and the rise of the skill-intensive sector. *Available at SSRN 5462774*, 2025.
- L. Wantchekon, M. Klačnja, and N. Novta. Education and human capital externalities: evidence from colonial benin. *The Quarterly Journal of Economics*, 130(2): 703–757, 2015.
- World Bank. Worldwide Governance Indicators, 2022. URL <https://www.worldbank.org/en/publication/worldwide-governance-indicators/interactive-data-access>.

Appendix

A Data Appendix

A.1 Additional details on the Blazing Furnace Anti-Corruption Campaign

An important change in Vietnam’s commitment to anti-corruption occurred in the middle of our study period in 2016 ([Hung, 2018](#)). Throughout the country’s reform era, multiple legislative efforts have been made to limit corruption, but most of these efforts fizzled or were limited in scope ([Malesky et al., 2019](#)). This changed in 2016 after several high-level corruption scandals that implicated critically important state-owned enterprises (SOEs) and several high-ranking officials. Immediately after the Party Congress in 2016, General Secretary Nguyen Phu Trong initiated an effort to root out corruption at its source. Trong coined the term “blazing furnace” at a meeting of the Central Steering Committee for Anti-Corruption in July 2017, arguing that corruption was constraining growth and promising to hold local and central officials accountable. Consequently, since 2016, over 200,000 party members, including two Presidents, the Speaker of the National Assembly, eight Politburo members, 39 Central Committee members, and 52 military and police generals, have been disciplined ([Giang, 2024](#)). Two other components of the blazing furnace efforts have been less attention-grabbing but are also consequential. First, Trong led a rewriting of VCP statutes to reduce malfeasance by limiting the permissible types of financial and business transactions. Second, the government pushed forward a series of e-governance reforms that have moved several public procedures and services online to streamline them and reduce opportunities for discretion by bureaucrats. The effort was intended

to combat corruption by reducing bottlenecks in official transactions and discretion by bureaucratic gatekeepers but has had limited take-up thus far (Malesky and Bui, 2024).

Surveys of firms and citizens indicate that the anti-corruption mobilization has been successful at reducing petty corruption. Today, 43% of businesses report that informal charges are common compared to 66% in 2016, and only 3.8% of businesses make bribe payments greater than 10% of annual revenue, compared to 9.1% in 2016.³⁴ Vietnamese citizens also report lower levels of bribery to local government in accessing public services and state employment and are more likely to believe the government is willing to combat corrupt activities.³⁵

It is possible that the scale and intensity of the anticorruption campaign may have moderated the impact of education on bribery. After 2016, citizens had more avenues for contesting and appealing corrupt activity and more avenues for avoiding bribery in daily transactions. If education increases awareness of these possibilities, we might expect a stronger negative relationship after 2016. We explore this possibility in more detail below.

A.2 The PAPI Dataset

To measure individual-level experiences with corruption we employ the Vietnamese Provincial Governance and Public Administration Performance Index (PAPI) annual survey from 2011 to 2022. The PAPI survey is administered by the United Nations Development Program (UNDP) in collaboration with the Centre for Community Sup-

³⁴Edmund Malesky, Pham Ngoc Thach, Truong Duc Trong, Phan Tuan Ngoc, Quynh Nguyen, *The Vietnam Provincial Competitiveness Index: Measuring Economic Governance for Private Sector Development, 2022 Final Report* (Hanoi, Vietnam: Chamber of Commerce and Industry and United States Agency for International Development, 2023): 71

³⁵CECODES, VVF, and UNDP “The 2022 Viet Nam Governance and Public Administration Performance Index (PAPI 2022): Measuring Citizens’ Experiences.” CECODES. Technical report (2023): 10

port and Development Studies (CECODES) and Vietnam Fatherland Front (VFF). Since its initiation in 2009, it has grown to be Vietnam’s largest public opinion survey with between 12,000 and 18,000 annual respondents.³⁶ In this project, we use a dataset of repeated cross-sections of individuals from within selected Vietnamese communes. While the communes are consistent over time between 2011 and 2022, the sample of individuals is refreshed every year.

The PAPI sampling design is ideal for our project because it includes a large set of representative samples of 320 districts,³⁷ including all 63 provincial capitals, that were surveyed every year between 2011 and 2022 and can be matched directly to the locations of university openings.³⁸ In 2011,³⁹ the PAPI research deployed a multi-level sampling approach to ensure representative samples of Vietnamese citizens over eighteen years old at the national level and for all 63 Vietnamese provinces and national-level cities. Units below the provincial administration level (district, commune, and village) were selected using probability proportion to size (PPS) sampling. Within each province, the capital district was always selected as a certainty unit, however, other districts were selected in a lottery weighted by the size of the district at the time of the 2010 Vietnamese Population Census. In provinces with over two million people, six districts were selected using PPS, while for provinces with under two million people, three districts were chosen. This same PPS approach is used to select capital and representative communes and villages within each district.

Once villages are selected, the PAPI research team travels to each village to ask for the most recent village household residency list, which includes all adults (over

³⁶Additional information on the methodology of the PAPI survey, dataset, and its usage can be found here: <https://papi.org.vn/eng/>

³⁷Out of a total of 703 districts

³⁸The PAPI survey was re-calibrated to match the 2019 population census and sampled a new set of communes, disrupting 50% of the village panels

³⁹In 2009 and 2010, PAPI only surveyed 3 and 30 provinces respectively, as it was refining its methodological approach

eighteen) in each village household. At this point, a simple random sample is used to select 20 households in 61 provinces and 40 households in the two metropolises of Hanoi and Ho Chi Minh City. The nearest birthday method (i.e. selection of the respondent with the birthday closest to the date of the survey) is used to provide an additional household-level randomization to ensure age and gender representativeness in the survey. The average response rate in our sample is 83% due to the highly reliable contact information provided by the village residence list.

Once an individual is selected, they are sent an invitation to travel to the village cultural center for an interview on a selected date.⁴⁰ Interviews within each village are staggered and take place in secluded locations within the cultural center to provide confidentiality, and a separate “expert survey” was designed to interview curious local government officials at a separate location away from the center to avoid creating anxiety among respondents about answering sensitive governance questions. Enumerator-answered control questions are available in the survey to track respondents’ level of comprehension, anxiety, and any salient disruptions.

CECODES deploys a survey research team of several hundred interviewers, who are selected from the province to ensure cultural, ethnic, and accent matches with the respondents. However, no interviewers are ever deployed in their home village. These survey teams are selected and trained by a group of survey managers from all 63 provinces, who travel to Hanoi each summer for a one-week training program on the sampling strategy, survey design, and questionnaire with research managers. At that time, complex and sensitive questions are explained and pre-scripted nudges are provided to aid respondents who struggle to comprehend questions.

The PAPI survey covers six governance modules including detailed batteries on

⁴⁰Culture centers were deemed more appropriate than within-home interviews because some households in mountainous or delta areas could be extremely difficult to reach and because poorer respondents were reluctant to allow enumerators into their homes.

corruption and quality of education. It also includes two modules on basic demographics (age, ethnicity, gender, party membership) and income and household assets. The survey, which since 2012 has been administered on a handheld tablet computer is about 54 pages long and, depending on the filter pattern, takes between 45 and 75 minutes to complete.

A.3 Data on University Openings

We manually collected data on the opening dates of all universities in Vietnam established before 2023. The data comes from a variety of official government documents and is an updated version of the university openings data used by [Vu \(2025\)](#).

A.4 Linking the PAPI Survey Data and the University Openings

The university openings data and the PAPI survey data are linked through administrative geocodes provided by the Vietnam General Statistics Office (GSO).

As administrative units in Vietnam change throughout the years, we have used the 2019 GSO codes for each district throughout the project to maintain consistency in identifying the locations of entities (universities and individuals).

A.5 Construction of Outcome variables

In this section, we explain how key outcome variables are defined. To measure the level of education, we created an indicator variable for some university education based on a survey question in PAPI, “What is your highest level of education?”. If the survey participant responds with “some university education”, “university education completed”, or “post-graduate degree”, the indicator variable is one, otherwise, it is 0.

For the main corruption measures, we created two indicator variables “personally affected” and “household affected” based on a survey question “Have you or anyone in your household been affected by an act of bribe-taking by a government official in the past year? Consider all types of bribe-giving ranging from giving a small bribe to a traffic official to giving a commission in order to get a government contract.” If the response is “not affected,” the indicator variables are zero. If “personally affected” is selected, the variable equals 1. If “personally affected” or “other family members affected,” the variable “household affected” equals 1.

We also leverage detailed questions in the PAPI survey to construct measures of corruption perceptions. Respondents choose “agree”, “somewhat agree”, or “disagree” for statements such as “people have to pay bribes in order to obtain a land title.” Besides land titles, respondents are also asked about perceptions of the need to pay bribes when obtaining a government job, receiving medical treatment at the district’s hospital, getting public primary school teachers to care more for one’s child, and when obtaining a construction permit, as well as to what extent government officials are diverting public funds for personal benefits. These measures are standardized and used in Figure 3 and Table 3. To create a summary index using these six perception measures, we follow [Anderson \(2008\)](#) to construct a weighted index using the inverted covariance matrix of all perception measures.

To test the hypothesis that more educated cohorts may denounce government officials since they have the capacity and understanding, we use this PAPI data survey question “Would you be willing to denounce the bribe if it was [amount]?”⁴¹ and “Did you or your family member attempt to denounce this act of corruption?” From these two questions, we create two indicator variables that are equal to 1 if the answer is

⁴¹In the survey, one bribe amount is chosen at random. Therefore, we additionally control for the bribe amount fixed effects for this outcome variable.

yes. Panel (a) of Figure 5 is answered by every respondent from 2015 to 2021, panel (b) only looks at the sub-sample of respondents from 2011 to 2012 who reported they were personally affected or their family members were affected by the act of bribe-taking, while panel (c) imputes the indicator variable to be 0 for everyone else who is unaffected, and hence looks at the entire population in the districts, unconditional on being affected or not.

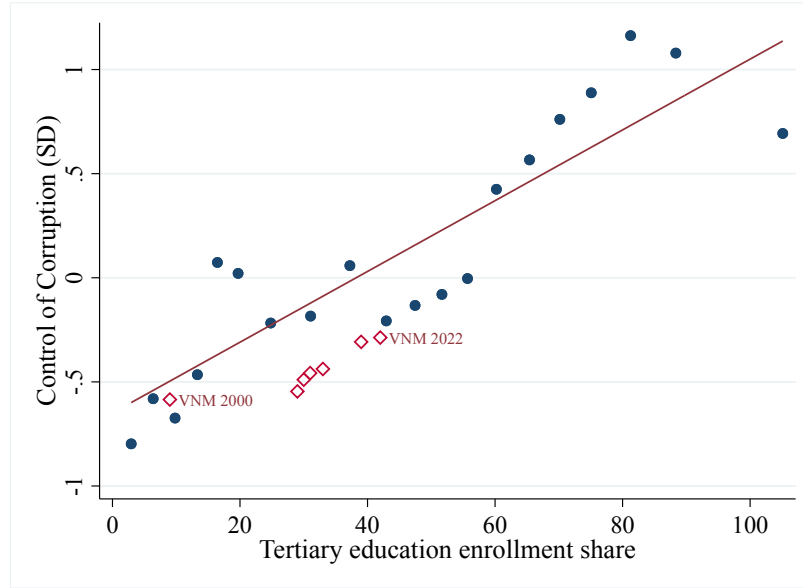
To examine the effect of university expansion on household income, we use the PAPI data’s survey question: “What is your household monthly income on average?” We take the mid-value of the income brackets selected by the survey participants. The outcome variable is denominated in a million Vietnamese dong (VND).

We also examine detailed interactions with government procedures and government-provided services to investigate the potential mechanism that more educated individuals may be more likely to have a certain interaction with government bureaucrats where bribes are common. We leverage questions in the PAPI data that ask specifically about the interaction with the government. For example, the PAPI survey asks the respondent to indicate “yes” or “not” to the statements “I/my family don’t have to pay bribes to obtain the service” and “Did you hire a middleman/facilitator in order to obtain this service for you?” when prompted with the most recent experience with the public certification service. The survey asks about four specific government procedures: applying for a public certification, applying for a construction permit, applying for a land use rights certificate, and other administrative procedures, such as obtaining state subsidies. The PAPI data does not ask about whether a middleman was used for government services of public hospitals and public primary schools. We then create two indicators that equal 1 if the survey participant paid extra informal charges for any of the 4 government procedures or bribed for any of the 2 government services. If not, the indicators will be zero. In panels (a) and (c) of Figure 4, we

focus on the sub-sample of the population that used any of the related government procedures or services. In panels (b) and (d), we impute the indicator variables to be 0 for everyone else who did not have any interaction with the government, and hence we look at the entire population in the districts, taking into account composition changes in the usage.

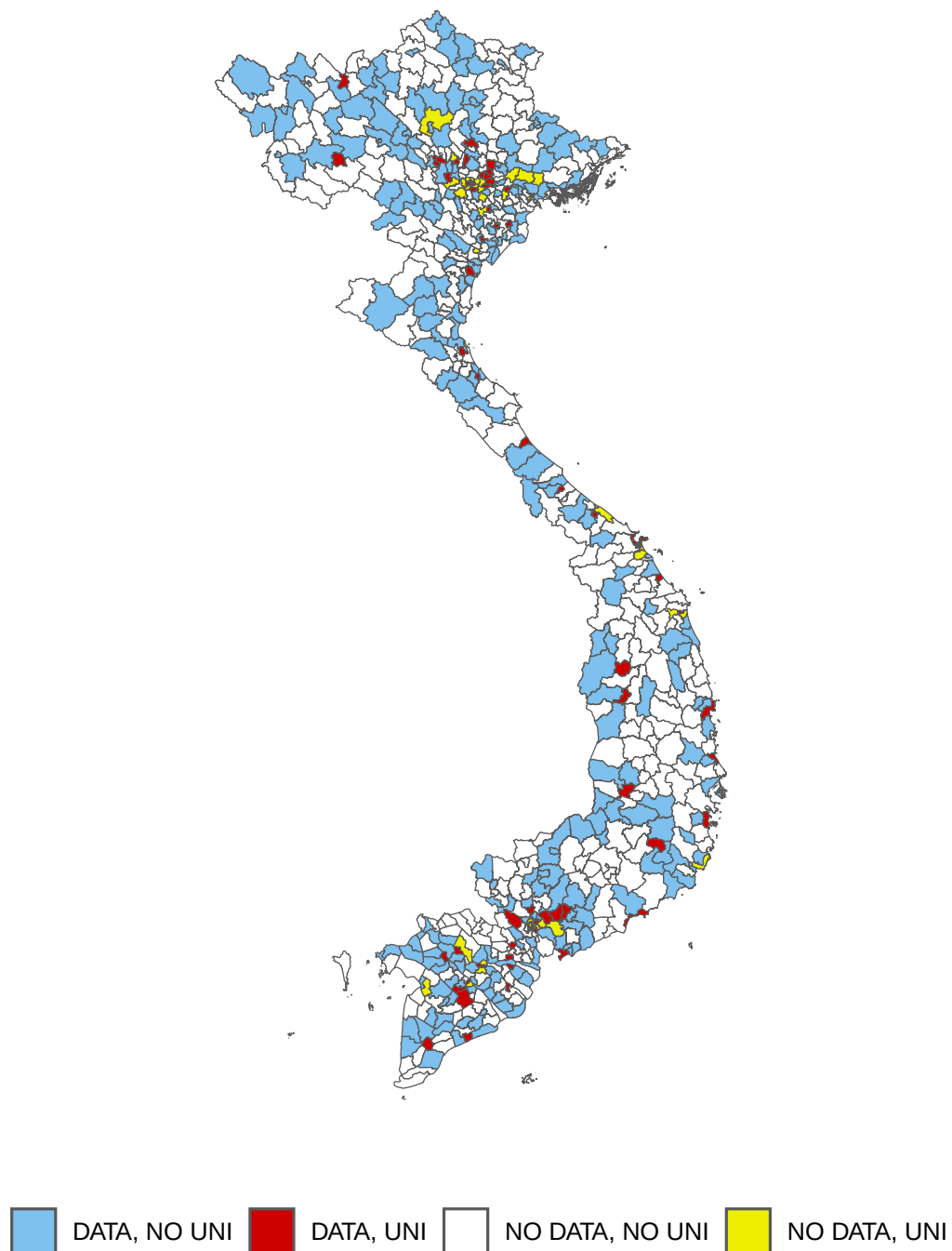
B Appendix Figures

Figure A.1: Cross-national correlation between corruption and higher education



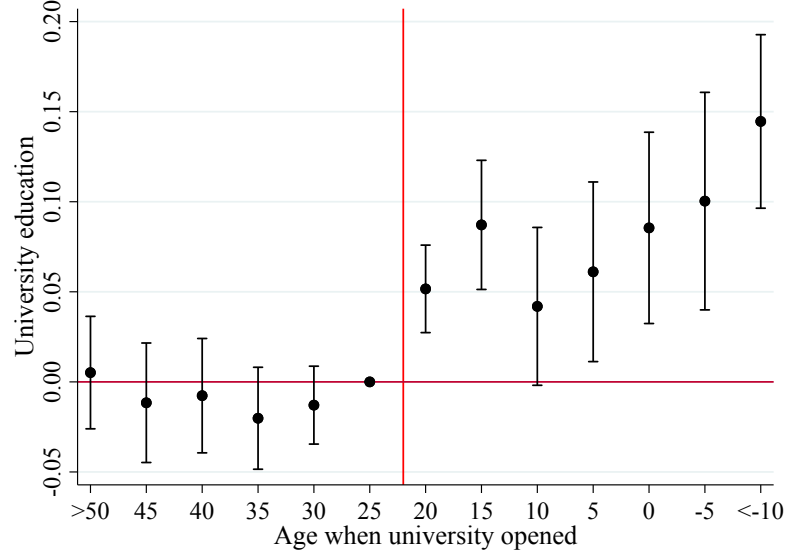
Note: This figure is a binned scatter plot based on two World Bank data series of tertiary education enrollment (% gross) and “Control of Corruption” for 180 countries from 2000 to 2022. The gross enrollment ratio is the ratio of total enrollment, regardless of age, to the population of the age group that officially corresponds to the level of education. The control of corruption variable captures perceptions of the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as “capture” of the state by elites and private interests, in units of standard deviations. The correlation is 0.502. The slope is 0.017, meaning that an increase of one percentage point in the share of tertiary enrollment is associated with a 0.017 standard deviation increase in the control of corruption measure (i.e., an improvement in the control of corruption). Raw observations for Vietnam are highlighted in red hollow diamonds and are not binned. See discussion in Section 1.

Figure A.2: Locations of New Universities Established During the Expansion



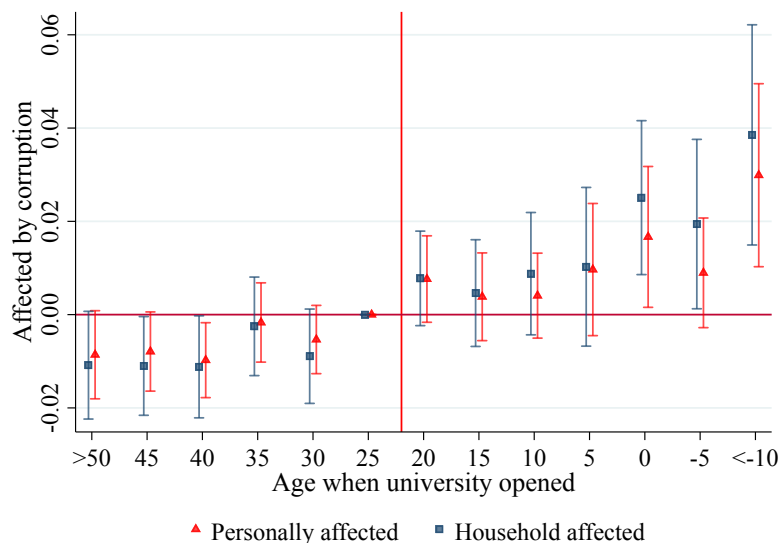
Note: This map of Vietnam displays four types of districts. Districts in blue are in the PAPI sample that never had a university open. Districts in red are in the PAPI sample and have a new university established. Districts in white are not in the PAPI sample and never had a university opened, and districts in yellow are not in the PAPI sample but had a university established. See discussion in Section 2.2.

Figure A.3: Effect of university expansion on respondents having a completed university degree



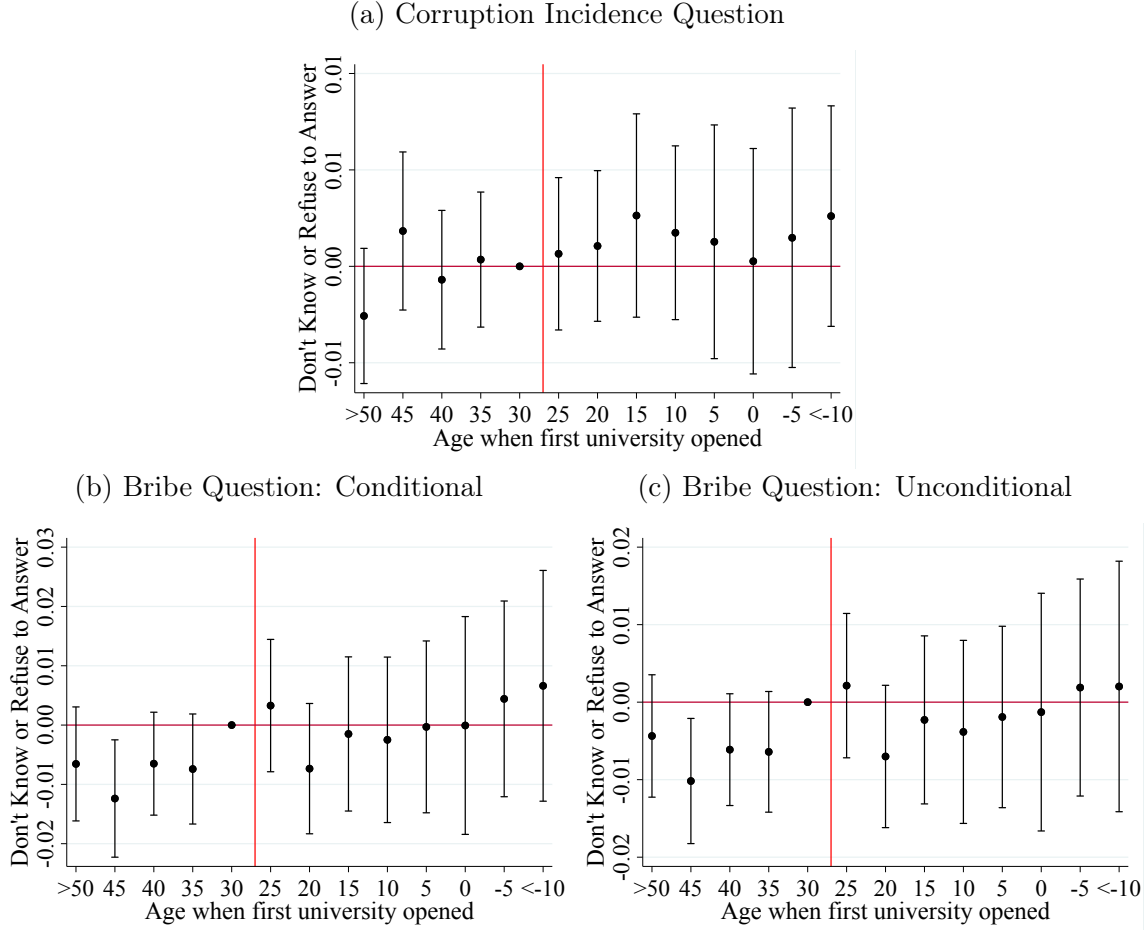
Note: Appendix Figure A.3 shows the effect of university expansion on the share of respondents who completed a university degree. We use Equation 1 to obtain the estimates for each cohort. The outcome variable is an indicator of respondents having a university degree. As the outcome is at the individual level, the reference age group is between 23 to 27. The mean of the outcome variable in districts that never had a university is 0.105. The number of observations is 171,162. 95% confidence intervals are constructed using standard errors clustered at the district level. Figure 1 shows the effect of university expansion on the share of receiving some university education. See discussion in Section 4.1.

Figure A.4: Effect of university expansion on whether respondent was affected by bribe-taking



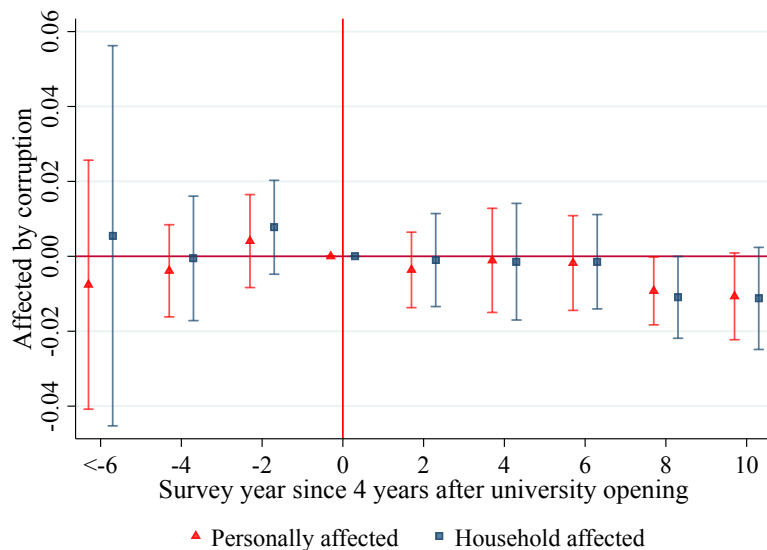
Note: Appendix Figure A.4 shows the effect of university expansion on the share having experienced an act of corruption in the previous year, using the reference age cohort from 23 to 27. Figure 2 uses an older reference group of 28 to 32 to account for the spillover effect from the younger members. We prefer the specification in Figure 2. For transparency, Appendix Figure A.4 is also presented. We use Equation 1 to obtain the estimates for each cohort (N=169,456). The outcome variable “personally affected” is an indicator of being personally affected by an act of bribe-taking, and “household affected” is an indicator of being personally affected or having a family member affected. The means of the outcome variables in districts that never had a university are 0.018 and 0.029 for personal and household experiences, respectively. The survey question is “Have you or anyone in your household been affected by an act of bribe-taking by a government official in the past year? Consider all types of bribe-giving, ranging from giving a small bribe to a traffic official to giving a commission in order to get a government contract.” 95% confidence intervals are constructed using standard errors clustered at the district level. See discussion in Section 4.2.

Figure A.5: Effect of university expansion on not answering corruption questions



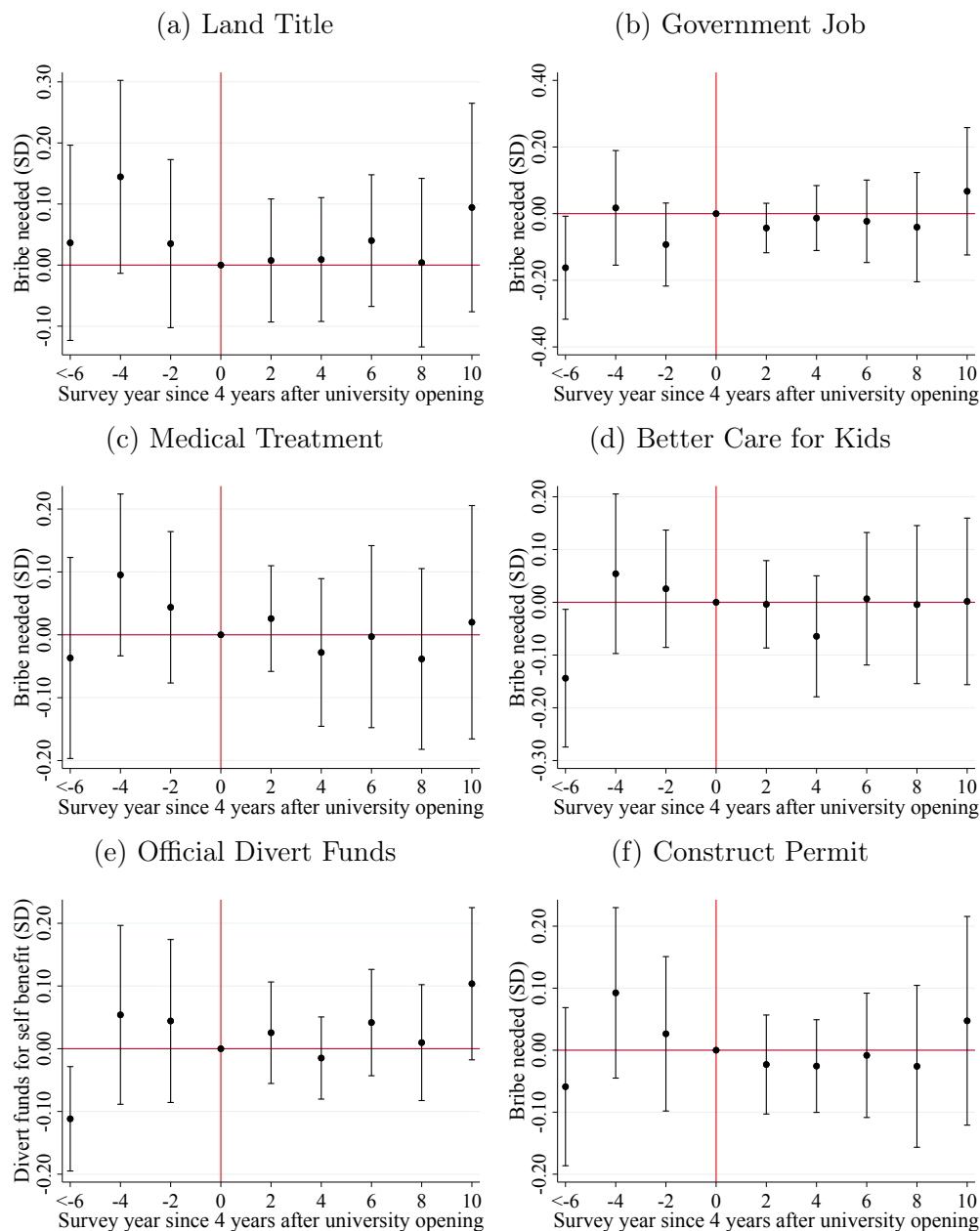
Note: Appendix Figure A.5 shows the effect of cohorts being exposed to a university opening on the share answering questions about corruption with “don’t know” or “refuse to answer”. The outcome variable is an indicator variable for whether any of the corruption questions were answered with “Don’t know” or “Refuse to answer.” The corruption question in panel (a) refers to the main corruption exposure question used in Figure 2. The bribe questions in panels (b) and (c) refer to questions about specific interactions with the government, for a procedure or service, used in Figure 4. Panels (b) is conditional on having had the interaction with the government, and Panel (c) includes all respondents. If a bribing question for any of the detailed interactions is answered with “don’t know” or “refuse to answer”, the indicator variables have the value 1. We use Equation 1 to obtain the event study estimates. The outcome variables are at the household level, so the reference age group is between 28 to 32. The mean of the outcome variable in districts that never had a university is 0.010 for panel a, 0.051 for panel b, and 0.042 for panel c. The number of observations is 171,162 for panel a, 139,881 for panel b, 171,162 for panel c. 95% confidence intervals are constructed with standard errors clustered at the district level. See discussion in Section 4.2.

Figure A.6: Effect of university expansion on whether households were affected by bribe-taking at the district level



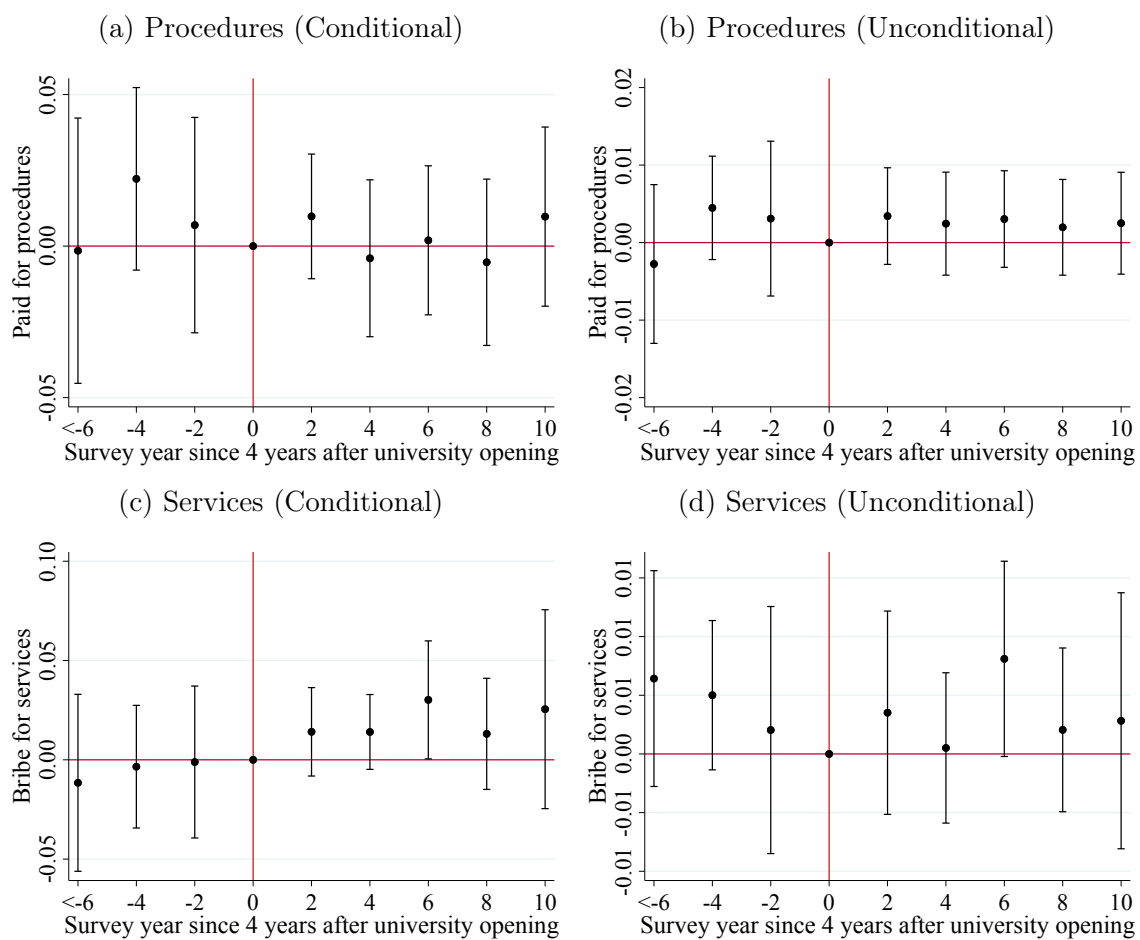
Note: Instead of using age-cohort difference-in-differences, Appendix Figure A.6 uses a standard difference-in-differences approach. It uses university openings that occur during the survey years, thereby replacing cohort group dummies with survey year dummies in Equation 1 and only keeping district and year fixed effects without age fixed effects (N=169,528). The outcome variables are the same as those in Figure 2. The share of those personally affected is 0.018 in districts that never had a university, and the share of households affected is 0.029. Survey participants who respond to the survey 4 years after the university opened form the reference group. 95% confidence intervals are constructed with standard errors clustered at the district level. The difference-in-differences estimates are reported in Appendix Table A.6. See discussion in Section 4.2.4.

Figure A.7: Effect of university expansion on respondents' perceptions of corruption at the district level



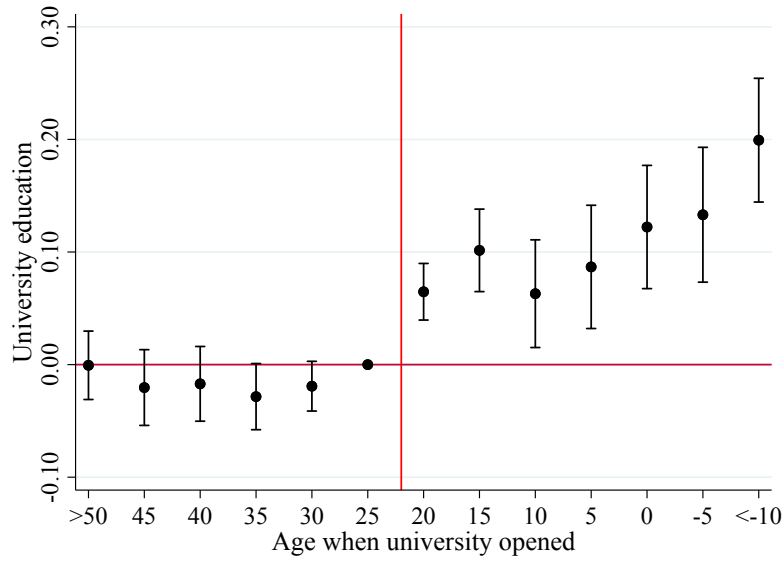
Note: Instead of using age-cohort difference-in-differences, Appendix Figure A.7 uses a standard difference-in-differences approach. It uses university openings that occur during the survey years, thereby replacing cohort group dummies with survey year dummies in Equation 1 and only keeping district and year fixed effects without age fixed effects. The outcome variables are the same as those in Figure 3. Survey participants who respond to the survey 4 years after the university opened form the reference group. 95% confidence intervals are constructed using standard errors clustered at the district level. The difference-in-differences estimates are reported in Appendix Table A.7. See discussion in Section 4.2.4.

Figure A.8: Effect of university expansion on informal payments during interactions with government at the district level



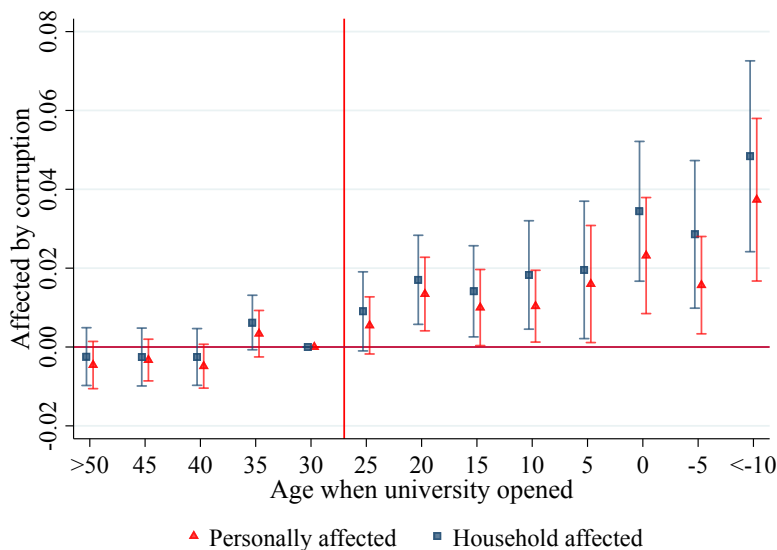
Note: Instead of using age-cohort difference-in-differences, Appendix Figure A.8 uses a standard difference-in-differences approach. It uses university openings that occur during the survey years, thereby replacing cohort group dummies with survey year dummies in Equation 1 and only keeping district and year fixed effects without age fixed effects. It shows the effect of cohorts being exposed to a university opening on households making informal payments at the district level. The outcome variables are the same as those in Figure 4. Panels (a) and (c) are conditional on having had the interaction ($N=100,473$ and $N=85,177$, respectively). Panels (b) and (d) use all observations ($N=171,243$). The means of the outcome variable in districts that never have a university are 0.105, 0.023, 0.092, and 0.028 respectively for panels (a), (b), (c), and (d). Survey participants who respond to the survey 4 years after the university opened form the reference group. 95% confidence intervals are constructed using standard errors clustered at the district level. The difference-in-differences estimates are reported in Appendix Table A.8. See discussion in Section 4.2.4.

Figure A.9: Stacked regression estimating the effect of university expansion on respondents receiving university education



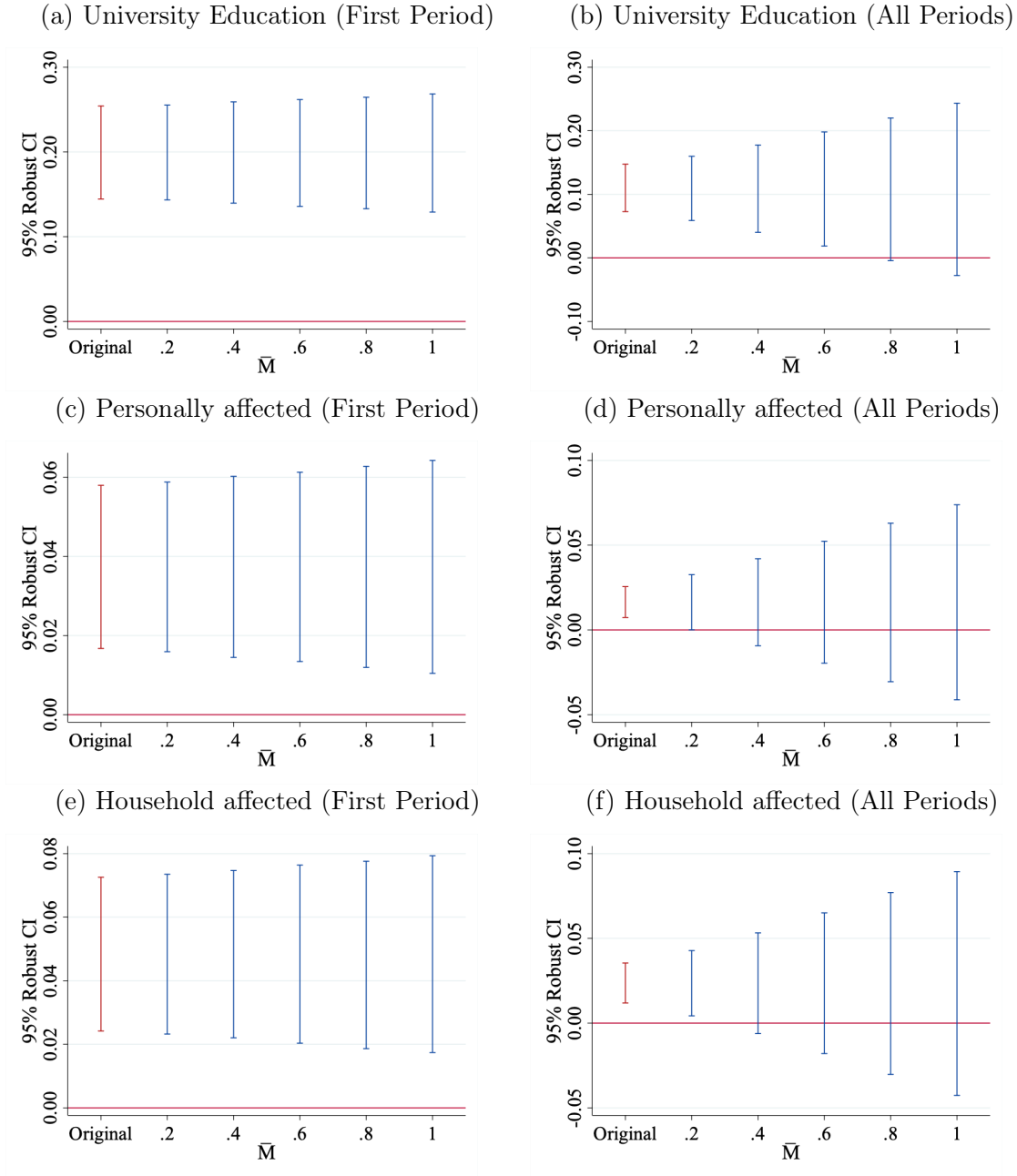
Note: We use the stacked difference-in-differences equation 3 to obtain the event study estimates. The outcome variable is an indicator of respondents being university-educated. This outcome is at the individual level, so the reference age group is between 23 to 27. The mean of the outcome variable in districts that never have a university is 0.112. The number of observations is 4,098,517. The question is, “What is your highest level of education?” 95% confidence intervals are constructed with standard errors clustered at the district level. The event study estimates from the baseline two-way fixed effects model are reported in Figure 1. See discussion in Section 4.3.

Figure A.10: Stacked regression estimating the effect of university expansion on whether the household was affected by bribe-taking



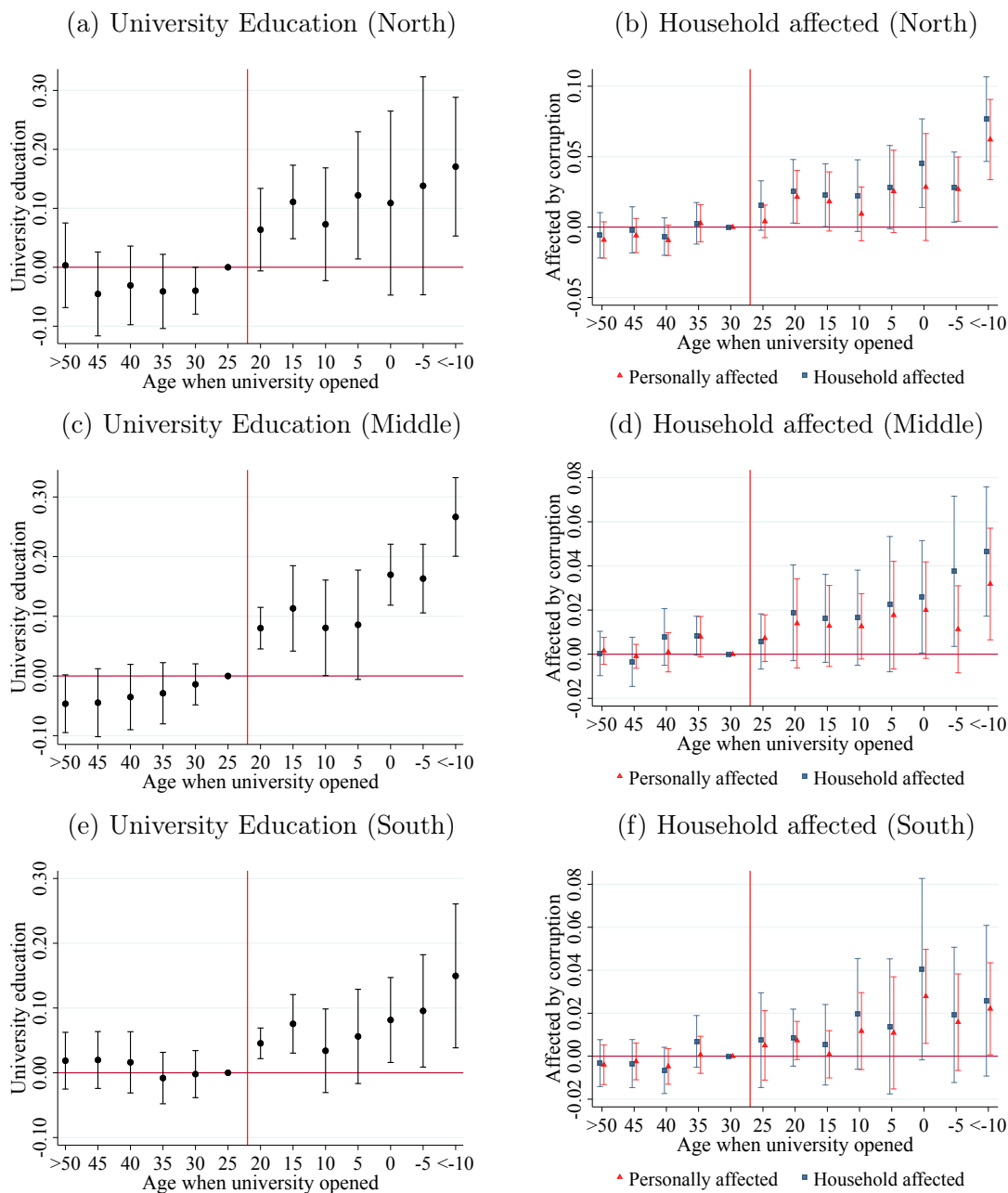
Note: We use the stacked difference-in-differences equation 3 to obtain the event study estimates (N=4,053,771). The outcome variable “personally affected” is an indicator of being personally affected by an act of bribe-taking and “household affected” is an indicator of being personally affected or having other family members affected. This outcome is at the household level, so the reference age group is between 28 to 32. The mean of the former in districts that never have a university is 0.018 and the mean of the latter is 0.029. The PAPI survey question is “Have you or anyone in your household been affected by an act of bribe-taking by a government official in the past year? Consider all types of bribe-giving, ranging from giving a small bribe to a traffic official to giving a commission in order to get a government contract.” 95% confidence intervals are constructed with standard errors clustered at the district level. The event study estimates from the baseline two-way fixed effects model are reported in Figure 2. See discussion in Section 4.3.

Figure A.11: Sensitivity to violations to the parallel trends assumption



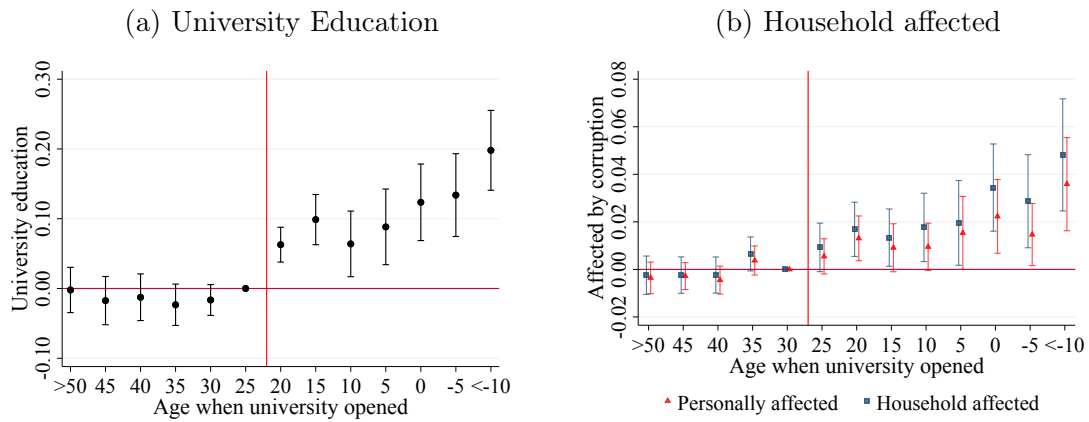
Note: This Figure shows confidence intervals robust to limited violations of the parallel trends assumption following [Rambachan and Roth \(2023\)](#). $\bar{M}$ indicates the extent of the post-treatment parallel trends assumption. $\bar{M} = 1$ allows for the post-treatment parallel trends assumption to be violated to the same extent in each period as the largest violation in the pre-treatment period. Using equation 3, panels (a), (c), (e) test the effect in the first post-period. Panels (b), (d), (f) test the average effect for all post-periods. See discussion in Section 4.3.

Figure A.12: Impact of university expansion on respondents receiving university education and households' corruption exposure at the regional level



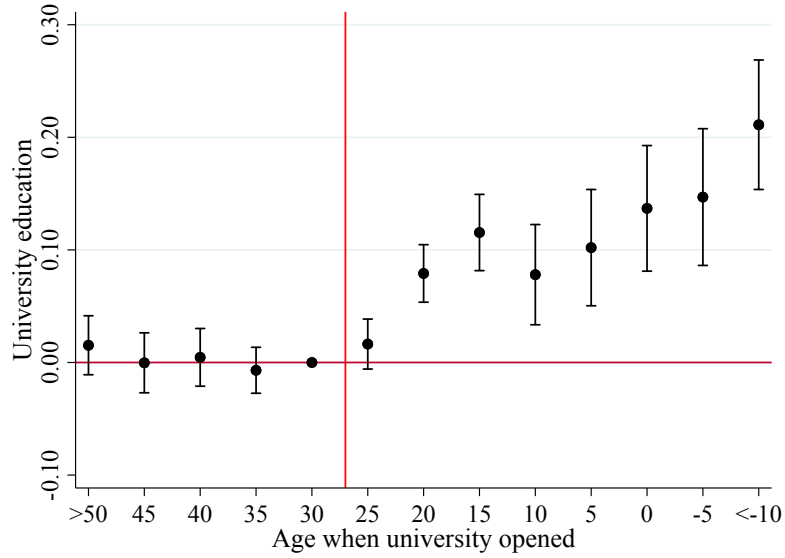
Note: Appendix Figure A.12 shows the impact of university expansion on university education and corruption exposure for different regions. Figure 1 and Figure 2 are replicated. For the north, middle, and south regions, the means of the education outcome in districts that never have a university opened are 0.156, 0.098, and 0.072. The shares of those personally affected are 0.023, 0.019, and 0.010, respectively. The shares of households affected are 0.037, 0.031, and 0.018, respectively. The numbers of observations for education outcomes are 64,510, 49,493, and 57,146. The numbers of observations for corruption exposure are 63,796, 48,982, and 56,666. See discussion in Section 4.3.

Figure A.13: Impact of university expansion on respondents receiving university education and households' corruption exposure with the additional control of provincial trends



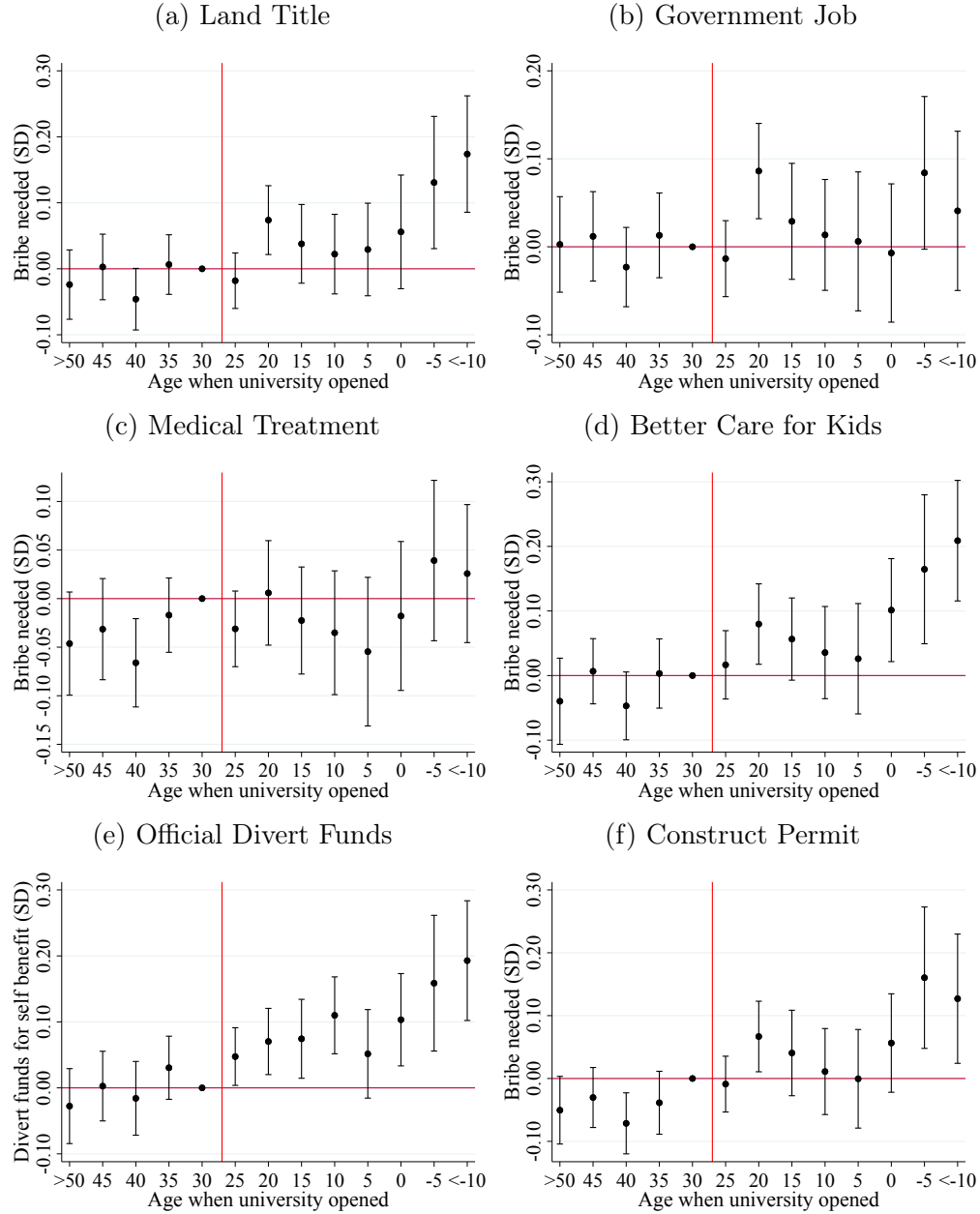
Note: Appendix Figure A.13 shows the impact of university expansion on university education and corruption exposure. Figure 1 and Figure 2 are replicated using Equation 1 with the additional control of provincial trends. The sample and the outcome variables are the same as those in Figure 1 and Figure 2. See discussion in Section 4.3.

Figure A.14: Effect of university expansion on university education, using the 28 to 32 age bin as reference group



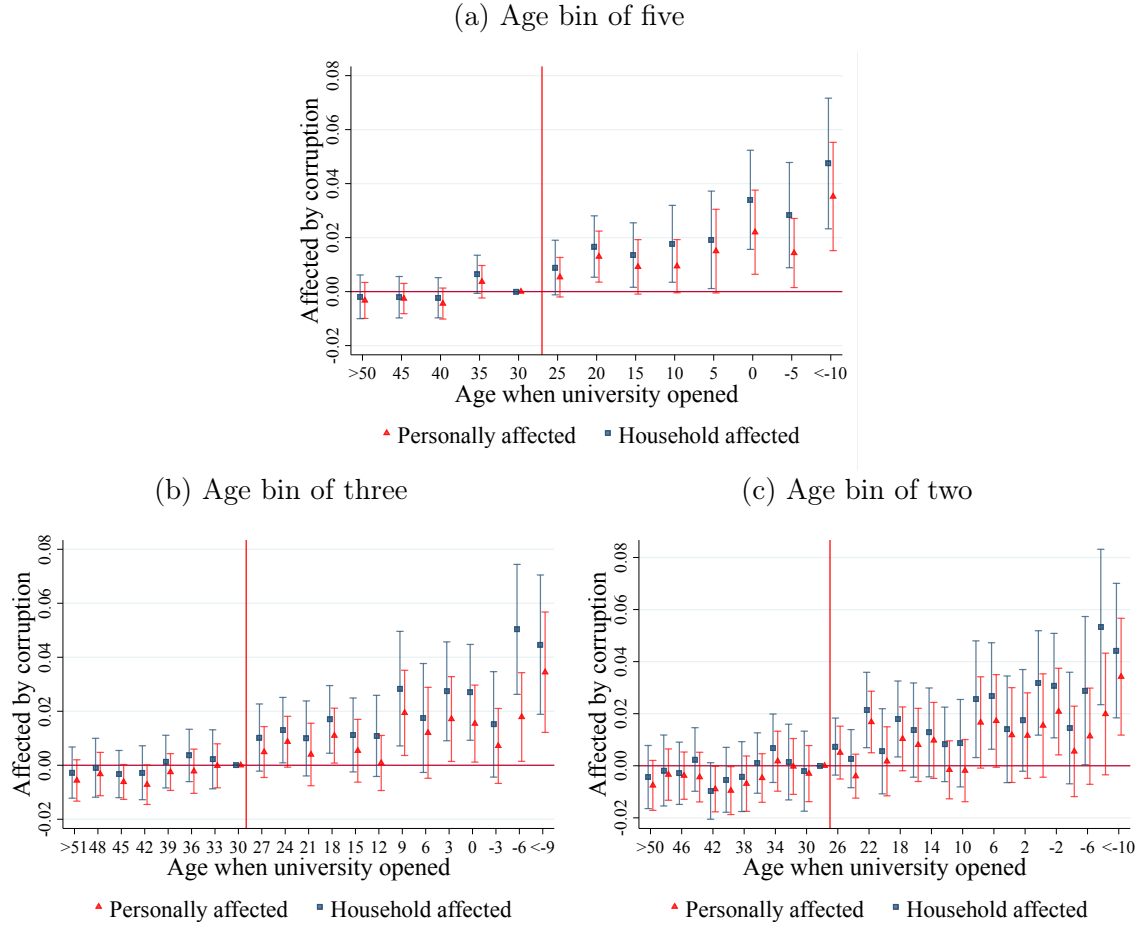
Note: Appendix Figure A.14 shows the effect of university expansion on respondents receiving some university education when the reference age group is between 28 to 32. We use Equation 1 to obtain the estimates for each cohort. The outcome variable is an indicator of the respondent being university-educated. As the outcome is at the individual level, our preferred specification is to use the 23 to 27 age group as the reference group, as is done in Figure 1. See discussion in Section 4.3.

Figure A.15: Effect of university expansion on perceptions of corruption, using the 28 to 32 age bin as reference group



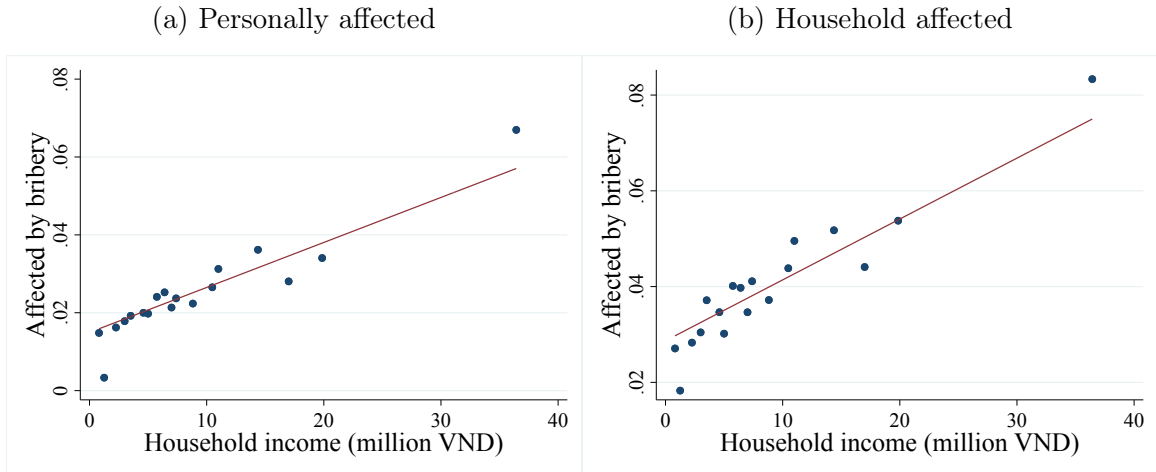
Note: Appendix Figure A.15 shows the effect of cohorts being exposed to a university opening on the perceptions of corruption, when the reference age group is between 28 to 32. The outcome variables are standardized measures of respondents' agreement with different statements related to corruption perceptions. As the outcome is at the individual level, our preferred specification is to use the 23 to 27 age group as the reference group, as is done in Figure 3. See discussion in Section 4.3.

Figure A.16: Different age bins for the effect of university expansion on households' probability of being affected by bribe-taking



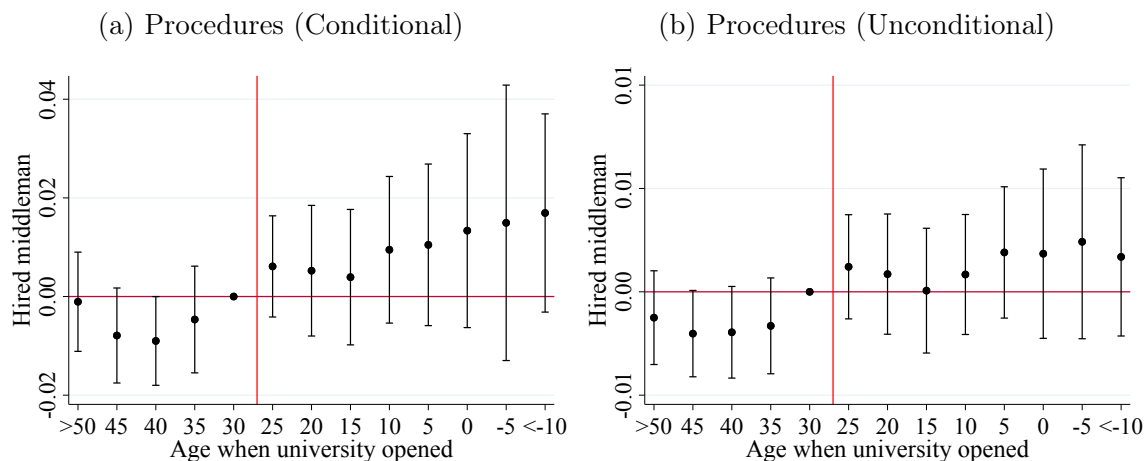
Note: We select different age bins to assess the robustness of our estimates for being affected by bribe-taking acts. In our main analysis, we use an age bin of five and omit the age group of 28-32 as the reference group (see discussion in Section 3). In panel b, we use an age bin of three and omit the age group of 28-30 as the reference group. In panel c, we use an age bin of two and omit the age group of 28-29 as the reference group. See discussion in Section 4.3.

Figure A.17: Relationship between being affected by bribe-taking and household income



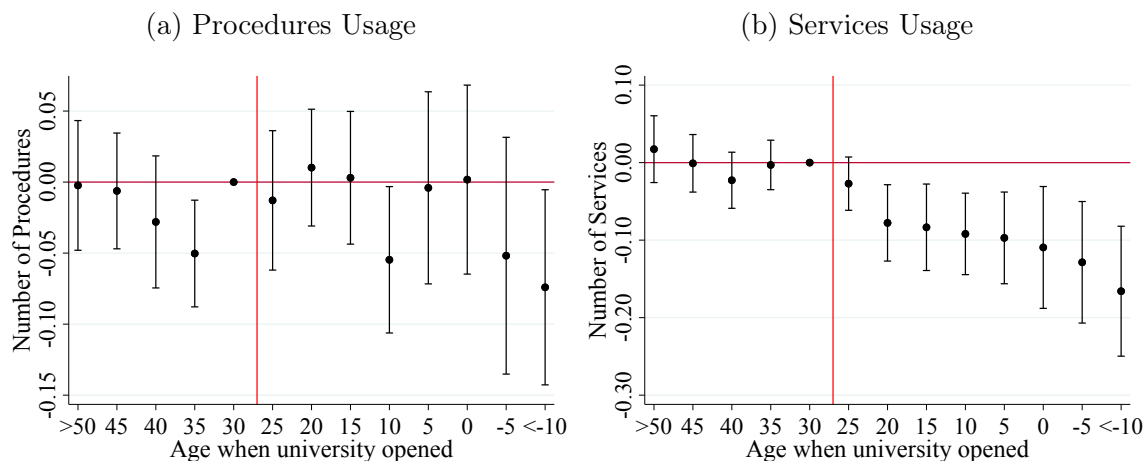
Note: The binscatter plots present the positive correlation between corruption exposure and household monthly income. These variables are created in the same manner as in Figure 2 and Figure 6, respectively. The OLS regression line was constructed using individual-level observations ($N=44,269$ and $N=44,269$). See discussion in Section 5.

Figure A.18: Effect of university expansion on paying a middleman to facilitate the household's interaction with a government procedure



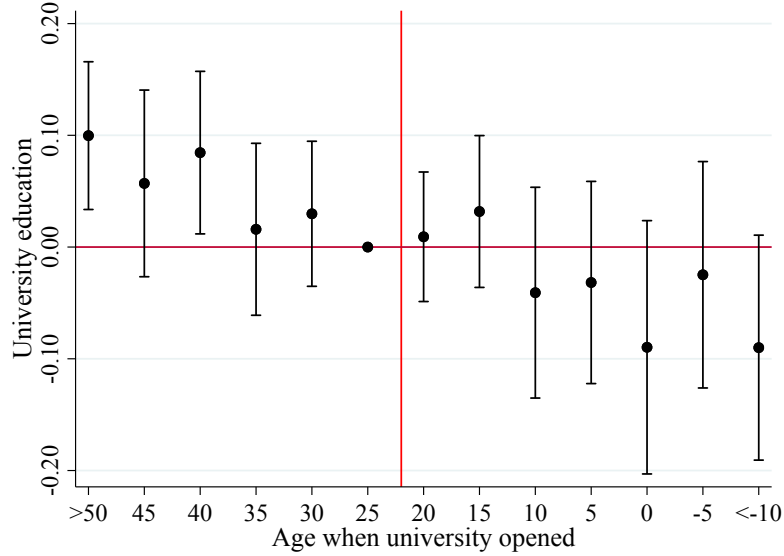
Note: We use Equation 1 to obtain the event study estimates for paying a middleman to facilitate a government procedure at the individual level. The outcome variable is an indicator variable of whether a middleman is hired for a government procedure. A government procedure could be an application for a certification, construction permit, land use rights, or other administrative procedures. This outcome is at the household level, so the reference age group is between 28 to 32. The mean of the outcome variable in districts that never had a university is 0.033 for panel a and 0.006 for panel b. The number of observations is 83,932 for panel a and 171,162 for panel b. 95% confidence intervals are constructed with standard errors clustered at the district level. The difference-in-differences point estimate for all cohorts is reported in Appendix Table A.15. See discussion in Section 5.

Figure A.19: Effect of university expansion on households' propensity to apply for government procedures and services



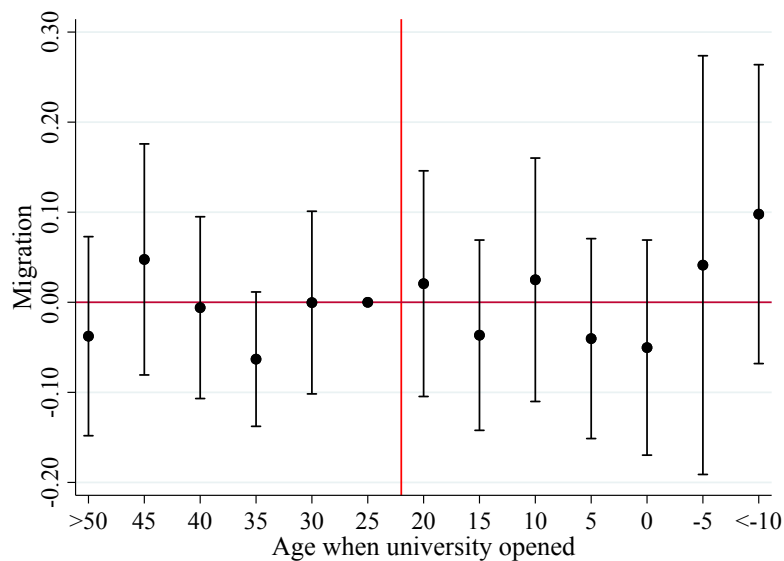
Note: Appendix Figure A.19 shows the impact of the university expansion on the propensity to apply for government procedures and services. We use Equation 1 to obtain the event study estimates for the number of times applying for government procedures and services at the individual level. The outcome variables of panel a and panel b are the number of procedures and services used, respectively. The procedures and services are the same as those in Figure 4. A government procedure could be an application for a certification, construction permit, land use rights, or other administrative procedures. The government services are public primary schools or public hospitals. This outcome is at the household level, so the reference age group is between 28 to 32. The mean of the outcome variable in districts that never had a university is 0.861 for panel a and 0.731 for panel b. The number of observations is 171,162. 95% confidence intervals are constructed with standard errors clustered at the district level. The difference-in-differences point estimate for all cohorts is reported in Appendix Table A.16. See discussion in Section 5.

Figure A.20: Effect of university expansion on the education level of the government officials



Note: Figure A.20 shows the effect of a cohort being exposed to a university opening on the share having a university education for a subsample of government officials. We use Equation 1 to obtain the estimates for each cohort. The outcome variable is an indicator of respondents being university-educated. This outcome is at the individual level, so the reference age group is between 23 to 27. The mean of the outcome variable in districts that never had a university is 0.536. The number of observations is 21,000 as we restrict to the subsample of the officials. 95% confidence intervals are constructed with standard errors clustered at the district level. See discussion in Section 5.

Figure A.21: Effect of university expansion on sample composition of migrants



Note: Figure A.21 shows the effect of being exposed to a university opening on the share of the migrant population. We use Equation 1 to obtain the estimates for each cohort. The outcome variable is an indicator of the respondent being a migrant. This outcome is at the individual level, so the reference age group is between 23 to 27. The mean of the outcome variable in districts that never have a university is 0.117. The number of observations is 171,162. 95% confidence intervals are constructed with standard errors clustered at the district level. See discussion in Section 5.

C Appendix Tables

Table A.1: District characteristics with or without a university opening

	Control Mean (1)	Standard Deviation (2)	Difference (3)
Capital district	0.08	0.28	0.57***
Urban	0.23	0.27	0.46***
Population	11531.34	3250.23	3218.66***
Size	63.21	68.65	-52.97***
Population/km ²	881.44	3382.85	3610.77***
Ethnic majority	0.72	0.34	0.23***
Distance to a special class city	182.30	131.17	0.04
District observations	Treatment: 69; Control: 251		

Note: Appendix Table A.1 compares district characteristics between districts with a university opening and control districts without one. Columns 1 and 2 report the means and standard deviations of the characteristics in the comparison districts, while Column 3 reports regression coefficients from regressions of each characteristic on an indicator for whether the district had a university opening. We use IPUMS data for districts from the 2019 population census published by the General Statistics Office (GSO) of Vietnam (Ruggles et al., 2025), as district codes are also standardized to the 2019 version. The capital district is an indicator of whether the district is located in the provincial capital. Cities in Vietnam are classified into four categories, and the special class is the category for the two largest cities, Ho Chi Minh City and Hanoi. We restrict the sample to districts for which we also have PAPI data and that are therefore part of our main analysis. See discussion in Section 2.2.

Table A.2: The effect of university expansion on respondents' corruption perceptions

	Perception Index		
	(1)	(2)	(3)
University Expansion	0.057*** (0.019)	0.058*** (0.019)	0.056*** (0.019)
Observations	163,744	163,744	163,744
Cluster No.	320	320	320
Adjusted R2	0.061	0.062	0.063
Mean of Never Treated	-0.026	-0.026	-0.026
District FE	✓	✓	✓
Year FE	✓	✓	✓
Age FE	✓	✓	✓
Demographic Controls		✓	✓
Full Controls			✓

Note: Appendix Table A.2 shows that the university expansion significantly increases the perception of corruption among younger cohorts. The perception index is an individual-level outcome which is the same as that in Table 3, capturing individual corruption perception towards “land title”, “government job”, “medical treatment”, “have children better attended”, “public funds”, and “construction permit”. We detail its construction in Appendix A.5. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 22 or below at the time when the university opened and 0 otherwise. Demographic controls include gender and ethnicity and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. See discussion in Section 4.2.

Table A.3: The effect of university expansion on households paying informal charges

	Procedure			Service		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Conditional						
University	0.0033	0.0042	0.0032	0.0018	0.0023	0.0021
Expansion	(0.0050)	(0.0050)	(0.0050)	(0.0059)	(0.0059)	(0.0058)
Observations	100,434	100,434	100,434	85,149	85,149	85,149
Cluster No.	320	320	320	320	320	320
Adjusted R2	0.014	0.015	0.017	0.021	0.021	0.022
Never T. Mean	0.105	0.105	0.105	0.092	0.092	0.092
Panel B: Unconditional						
University	0.0012	0.0012	0.0012	-0.0001	-0.0000	-0.0000
Expansion	(0.0014)	(0.0014)	(0.0014)	(0.0023)	(0.0023)	(0.0023)
Observations	171,162	171,162	171,162	171,162	171,162	171,162
Cluster No.	320	320	320	320	320	320
Adjusted R2	0.013	0.013	0.013	0.027	0.027	0.027
Never T. Mean	0.023	0.023	0.023	0.028	0.028	0.028
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Age FE	✓	✓	✓	✓	✓	✓
Demo. Controls		✓	✓		✓	✓
Full Controls			✓			✓

Note: Appendix Table A.3 shows the effect of university expansion on households of the younger cohort paying informal charges for government procedures or services. Panel (a) is conditional on using the government procedures or services, and panel (b) does not condition on the usage. The household-level outcome variables are the same as those in Figure 4. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 27 or below at the time when the university opened and 0 otherwise. Demographic controls include gender and ethnicity, and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. Appendix Table A.4 shows the effect on informal payments for government procedures excluding payment to the middleman. See discussion in Section 4.2.

Table A.4: The effect of university expansion on payments to officials for government procedures (excluding middlemen)

	Conditional			Unconditional		
	(1)	(2)	(3)	(4)	(5)	(6)
University Expansion	-0.0002 (0.0045)	0.0008 (0.0045)	-0.0002 (0.0045)	-0.0001 (0.0010)	-0.0001 (0.0010)	-0.0002 (0.0010)
Observations	95,685	95,685	95,685	171,162	171,162	171,162
Cluster No.	320	320	320	320	320	320
Adjusted R2	0.019	0.020	0.022	0.013	0.013	0.013
Never T. Mean	0.089	0.089	0.089	0.018	0.018	0.018
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Age FE	✓	✓	✓	✓	✓	✓
Demo. Controls		✓	✓		✓	✓
Full Controls			✓			✓

Note: Appendix Table A.4 shows the effect of university expansion on households of the younger cohort paying informal charges for government procedures, after excluding payment to middleman. Columns 1 to 3 are conditional on using the government procedures, and columns 4 to 6 do not condition on the usage. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 27 or below at the time when the university opened and 0 otherwise. Demographic controls include gender and ethnicity, and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. See discussion in Section 4.2.

Table A.5: The effect of university expansion on the bribery measured by list experiments

	(1)	(2)	(3)
	Land	Medical	Kids
University Expansion	0.030 (0.065)	0.060 (0.035)	-0.003 (0.044)
Observations	12,712	55,007	27,841
Adjusted R2	0.038	0.063	0.101
Mean of Never treated	0.213	0.104	0.087
District FE	✓	✓	✓
Year FE	✓	✓	✓
Age FE	✓	✓	✓

Note: Appendix Table A.5 shows that the university expansion has no precise impact on bribe behaviors elicited through the list experiment. The list experiment elicits the number of listed action items that a person did while interacting with government procedures or services. In one of the two list versions, there is an additional item related to “informal payment” which is randomly shown to half of the respondents. As the respondents only report the total number of items, they do not need to admit to paying a bribe. Although the list experiment has the benefit of reducing social desirability bias, the sample is restricted to individuals who had specific government interactions. The interaction with the government is at the household level. The specific interactions are applying for land use rights, or using government services such as public primary schools and public hospitals. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 27 or below at the time when the university opened and 0 otherwise. Standard errors are clustered at the district level. See discussion in Section 4.2.

Table A.6: The effect of university expansion on whether the household was affected by bribe-taking at the district level

	Personally Affected			Household Affected		
	(1)	(2)	(3)	(4)	(5)	(6)
University Expansion	-0.0005 (0.0045)	-0.0006 (0.0045)	-0.0016 (0.0044)	-0.0015 (0.0054)	-0.0016 (0.0055)	-0.0030 (0.0053)
Observations	169,528	169,528	169,528	169,528	169,528	169,528
Cluster No.	320	320	320	320	320	320
Adjusted R2	0.010	0.012	0.013	0.013	0.014	0.015
Never T. Mean	0.018	0.018	0.018	0.029	0.029	0.029
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Demo. Controls		✓	✓		✓	✓
Full Controls			✓			✓

Note: Appendix Table A.6 shows the district-level effect of exposure to the university expansion on the household’s probability of experiencing an act of bribe-taking. The outcome variable “personally affected” is an indicator of being personally affected by an act of bribe-taking, and “household affected” is an indicator of being personally affected or having a family member affected. The outcome variable is the same as that in Figure 2. Instead of using age-cohort difference-in-differences, Appendix Table A.6 uses a standard difference-in-differences approach. It uses university openings that occur during the survey years. We combine survey year dummies into an independent variable “University Expansion” which equals 1 if the district is surveyed 4 years after university opening with a survey entry in the pre-period, and 0 otherwise. “Demographic controls” include gender and ethnicity. “Full controls” additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. Event study coefficients are reported in Appendix Figure A.6. See discussion in Section 4.2.4.

Table A.7: The effect of university expansion on respondents' corruption perceptions at the district level

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Land	Govt Job	Medical	Kids	Funds	Permit	Index
University Expansion	-0.023 (0.039)	0.025 (0.063)	-0.027 (0.048)	0.004 (0.048)	-0.034 (0.037)	-0.017 (0.040)	-0.003 (0.055)
Observations	140,240	138,444	150,000	149,631	135,889	138,628	163,805
Cluster No.	320	320	320	320	320	320	320
Adjusted R2	0.034	0.111	0.069	0.038	0.018	0.047	0.057
Mean of Never Treated	-0.006	-0.032	-0.009	-0.046	0.009	-0.053	-0.026
District FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓

Note: Appendix Table A.7 shows the district-level effect of university expansion on respondents' perception of several corrupt behaviors, such as "people have to bribe to obtain a land title", or "for a government job", "to receive medical treatment in the public hospitals", or "have their children better attended", or "officials divert funds for their personal benefit", or to "obtain construction permit". Respondents choose an option among "agree", "somewhat agree", and "disagree". We standardize the outcome variables to have a mean of zero and a standard deviation of one. The overall index in column 7 is a standardized weighted average of these standardized components, which also increases as a result of university expansion (see Appendix A.5 for details of the outcome construction). These outcome variables are the same as those in Figure 3. Instead of using age-cohort difference-in-differences, Appendix Table A.7 uses a standard difference-in-differences approach. It uses university openings that occur during the survey years. We combine survey year dummies into an independent variable "University Expansion" which equals 1 if the district is surveyed 4 years after university opening with a survey entry in the pre-period, and 0 otherwise. Standard errors are clustered at the district level. For event study graphs, see Appendix Figure A.7. See discussion in Section 4.2.4.

Table A.8: The effect of university expansion on households paying informal charges at the district level

	Procedure			Service		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Conditional						
University Expansion	-0.0080 (0.0090)	-0.0076 (0.0091)	-0.0093 (0.0092)	0.0180 (0.0135)	0.0182 (0.0136)	0.0184 (0.0137)
Observations	100,473	100,473	100,473	85,177	85,177	85,177
Cluster No.	320	320	320	320	320	320
Adjusted R2	0.014	0.015	0.017	0.017	0.018	0.018
Never T. Mean	0.105	0.105	0.105	0.092	0.092	0.092
Panel B: Unconditional						
University Expansion	-0.0007 (0.0026)	-0.0006 (0.0026)	-0.0008 (0.0026)	0.0023 (0.0038)	0.0024 (0.0038)	0.0020 (0.0036)
Observations	171,243	171,243	171,243	171,243	171,243	171,243
Cluster No.	320	320	320	320	320	320
Adjusted R2	0.012	0.012	0.012	0.020	0.021	0.021
Never T. Mean	0.023	0.023	0.023	0.028	0.028	0.028
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Demo. Controls		✓	✓		✓	✓
Full Controls			✓			✓

Note: Appendix Table A.8 shows the district-level effect of university expansion on households of the younger cohort paying informal charges for government procedures or services. Panel (a) is conditional on using the government procedures or services, and panel (b) does not condition on the usage. The household-level outcome variables are the same as those in Figure 4. Instead of using age-cohort difference-in-differences, Appendix Table A.8 uses a standard difference-in-differences approach. It uses university openings that occur during the survey years. We combine survey year dummies into an independent variable “University Expansion” which equals 1 if the district is surveyed 4 years after university opening with a survey entry in the pre-period, and 0 otherwise. Demographic controls include gender and ethnicity, and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. Event study coefficients are reported in Appendix Figure A.8. See discussion in Section 4.2.4.

Table A.9: The effect of university expansion on respondents receiving some university education, with Conley standard errors

	University Education		
	(1)	(2)	(3)
University Expansion	0.0875*** (0.0169)	0.0846*** (0.0161)	0.0939*** (0.0128)
Observations	171,162	171,162	171,162
Adjusted R ²	0.110	0.116	0.314
Mean of Never Treated	0.112	0.112	0.112
Conley SE	✓	✓	✓
District FE	✓	✓	✓
Year FE	✓	✓	✓
Age FE	✓	✓	✓
Demographic Controls		✓	✓
Full Controls			✓

Note: Appendix Table A.9 shows the robustness of the result in Table 1 with Conley standard errors that adjust for spatial correlations. The distance cutoff used is 300km. Demographic controls include gender and ethnicity and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. See discussion in section 4.3.

Table A.10: The effect of university expansion on households' corruption exposure, with Conley standard errors

	Personally Affected			Household Affected		
	(1)	(2)	(3)	(4)	(5)	(6)
University Expansion	0.0093*** (0.0034)	0.0090*** (0.0034)	0.0089*** (0.0034)	0.0121*** (0.0044)	0.0117*** (0.0044)	0.0117** (0.0054)
Observations	169,456	169,456	169,456	169,456	169,456	169,456
Adjusted R ²	0.016	0.018	0.018	0.021	0.022	0.022
Never T. Mean	0.018	0.018	0.018	0.029	0.029	0.029
Conley SE	✓	✓	✓	✓	✓	✓
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Age FE	✓	✓	✓	✓	✓	✓
Demo. Controls		✓	✓		✓	✓
Full Controls			✓			✓

Note: Appendix Table A.10 shows the robustness of the result in Table 2 with Conley standard errors that adjust for spatial correlations. The distance cutoff used is 300km. Demographic controls include gender and ethnicity and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. See discussion in section 4.3.

Table A.11: The effect of university expansion on respondents' knowledge about anti-corruption legislation

	Knowledge of Anti-corruption		
	(1)	(2)	(3)
University Expansion	0.013 (0.009)	0.006 (0.009)	0.014* (0.008)
Observations	170,596	170,596	170,596
Cluster No.	320	320	320
Adjusted R2	0.046	0.075	0.129
Mean of Never Treated	0.484	0.484	0.484
District FE	✓	✓	✓
Year FE	✓	✓	✓
Age FE	✓	✓	✓
Demographic Controls		✓	✓
Full Controls			✓

Note: Appendix Table A.11 shows that university expansion marginally increases the knowledge of anti-corruption legislation. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 22 or below at the time when the university opened and 0 otherwise. The individual-level outcome variable is an indicator that is equal to one if the survey respondent heard about the Law on Corruption Prevention – Anti Corruption and zero otherwise. Demographic controls include gender and ethnicity and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. See discussion in Section 5.

Table A.12: The effect of university expansion on respondents' denouncement of corrupt government officials

	Denounce		
	(1)	(2)	(3)
University Expansion	-0.0176 (0.0108)	-0.0189* (0.0108)	-0.0186* (0.0107)
Observations	93,720	93,720	93,720
Cluster No.	314	314	314
Adjusted R2	0.147	0.151	0.151
Mean of Never Treated	0.598	0.598	0.598
District FE	✓	✓	✓
Year FE	✓	✓	✓
Age FE	✓	✓	✓
Demographic Controls		✓	✓
Full Controls			✓

Note: Table A.12 shows that the university expansion has no or a marginally negative impact on the hypothetical denunciation actions of the exposed younger cohorts. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 22 or below at the time when the university opened and 0 otherwise. The survey question “If you found yourself the victim of a bribe request, [w]ould you be willing to denounce the bribe if it was [amount]?” is used. Only for this outcome, we additionally control for the hypothetical bribe amount fixed effects. The outcome variable is at the individual level and is the same as that in panel (a) of Figure 5. Demographic controls include gender and ethnicity and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. See discussion in Section 5.

Table A.13: The effect of university expansion on households' denunciation of government officials

	Denounce		
	(1)	(2)	(3)
Panel A: Conditional			
University Expansion	-0.0124 (0.0121)	-0.0121 (0.0121)	-0.0125 (0.0119)
Observations	5,517	5,517	5,517
Cluster No.	274	274	274
Adjusted R2	0.027	0.028	0.028
Mean of Never Treated	0.069	0.069	0.069
Panel B: Unconditional			
University Expansion	-0.0002 (0.0006)	-0.0002 (0.0006)	-0.0002 (0.0006)
Observations	171,162	171,162	171,162
Cluster No.	320	320	320
Adjusted R2	0.000	0.000	0.000
Mean of Never Treated	0.002	0.002	0.002
District FE	✓	✓	✓
Year FE	✓	✓	✓
Age FE	✓	✓	✓
Demographic Controls		✓	✓
Full Controls			✓

Note: Appendix Table A.13 shows that the university expansion has no significant impact on the denunciation actions of the exposed younger cohorts. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 27 or below at the time when the university opened and 0 otherwise. Panel (a) is conditional on the household being affected by a bribe-taking act, and panel (b) does not condition on being affected or not. The household-level outcome variables are the same as those in panels (b) and (c) of Figure 5. Demographic controls include gender and ethnicity and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. See discussion in Section 5.

Table A.14: The effect of university expansion on household income

	Household Income		
	(1)	(2)	(3)
University Expansion	1.315*** (0.326)	1.214*** (0.328)	1.272*** (0.331)
Observations	44,529	44,529	44,529
Cluster No.	208	208	208
Adjusted R2	0.168	0.177	0.200
Mean of Never Treated	7.535	7.535	7.535
District FE	✓	✓	✓
Year FE	✓	✓	✓
Age FE	✓	✓	✓
Demographic Controls		✓	✓
Full Controls			✓

Note: Appendix Table A.14 shows that the university expansion increases the average household monthly income by 1.31 million VND. This is a 17.4% increase, relative to the mean of districts that never had a university opened. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 27 or below at the time when the university opened and 0 otherwise. The household-level outcome variable is the same as that in Figure 6. Demographic controls include gender and ethnicity and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. See discussion in Section 5.

Table A.15: The effect of university expansion on hiring a middleman to facilitate a household's interaction with a government procedure

	Payment to Middleman		
	(1)	(2)	(3)
Panel A: Conditional			
University Expansion	0.0100*** (0.0035)	0.0102*** (0.0035)	0.0100*** (0.0035)
Observations	83,932	83,932	83,932
Cluster No.	209	209	209
Adjusted R2	0.009	0.009	0.009
Mean of Never Treated	0.033	0.033	0.033
Panel B: Unconditional			
University Expansion	0.0022*** (0.0008)	0.0021** (0.0008)	0.0022*** (0.0008)
Observations	171,162	171,162	171,162
Cluster No.	320	320	320
Adjusted R2	0.011	0.011	0.011
Mean of Never Treated	0.006	0.006	0.006
District FE	✓	✓	✓
Year FE	✓	✓	✓
Age FE	✓	✓	✓
Demographic Controls		✓	✓
Full Controls			✓

Note: Appendix Table A.15 Panel A shows that the university expansion increases the likelihood of hiring a middleman by 30.2% to facilitate a government procedure, conditional on the household having interacted with the government procedure. Panel B reports the unconditional version. The household-level outcome variable is the same as that in Figure A.18. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 27 or below at the time when the university opened and 0 otherwise. Standard errors are clustered at the district level. See discussion in Section 5.

Table A.16: The effect of university expansion on the number of government procedures and services households applied to

	Number of Procedure			Number of Service		
	(1)	(2)	(3)	(4)	(5)	(6)
University Expansion	0.0148 (0.0160)	0.0085 (0.0163)	0.0168 (0.0164)	-0.0561*** (0.0158)	-0.0563*** (0.0158)	-0.0538*** (0.0157)
Observations	171,162	171,162	171,162	171,162	171,162	171,162
Cluster No.	320	320	320	320	320	320
Adjusted R2	0.045	0.055	0.070	0.069	0.069	0.071
Never T. Mean	0.861	0.861	0.861	0.731	0.731	0.731
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Age FE	✓	✓	✓	✓	✓	✓
Demo. Controls		✓	✓		✓	✓
Full Controls			✓			✓

Note: Appendix Table A.16 shows that the university expansion does not increase the usage of government procedures significantly and decreases the usage of public hospitals and primary schools. The outcome variables of columns 1-3 and columns 4-6 are the numbers of procedures and services used respectively. The procedures and services are the same as those in Figure 4. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 27 or below at the time when the university opened and 0 otherwise. The household-level outcome variables are the same as those in Appendix Figure A.19. Demographic controls include gender and ethnicity and full controls additionally include an indicator for party membership, an indicator for working for the government, and an indicator for being a migrant. Standard errors are clustered at the district level. See discussion in Section 5.

Table A.17: The effect of university expansion on education, knowledge of anti-corruption legislation, propensity to denounce corrupt officials, and corruption experiences, before and after the anti-corruption campaign

	(1)	(2)	(3)	(4)	(5)
	Uni Edu	Knowledge	Denounce	Personal	Household
University Expansion	0.0749*** (0.0161)	-0.0034 (0.0130)	-0.0467* (0.0245)	0.0075 (0.0046)	0.0110* (0.0061)
University Expansion $\times$ Post 2016	0.0124 (0.0156)	0.0240* (0.0143)	0.0586** (0.0296)	0.0027 (0.0051)	0.0011 (0.0067)
Observations	171,159	170,593	5,506	169,453	169,453
Cluster No.	320	320	274	320	320
Adjusted R2	0.114	0.049	0.037	0.017	0.023
Mean of Never Treated	0.112	0.484	0.069	0.018	0.029
District FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Age FE	✓	✓	✓	✓	✓

Note: Appendix Table A.17 shows that the effects of the university expansion, before and after the start of the 2016 anti-corruption campaign. Outcome variables are some university education (Column 1), knowledge about the Law on Corruption Prevention (Column 2), the propensity to denounce corrupt officials (Column 3), and our main corruption exposure measures at the personal and household levels (columns 4 and 5). The outcome variables in columns 1, 2, 3, 4, and 5 are the same as those in Figure 1, Appendix Table A.11, Figure 5 panel (b), and Figure 2. The outcome variables are individual-level for columns 1 and 2, and household-level for columns 3 to 5. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 22 or below for individual-level outcomes in columns 1 to 2 and 27 or below for household-level outcomes of columns 3 to 5 at the time when the university opened and 0 otherwise. We further interact this indicator variable and fixed effects with the indicator variable, which equals 1 if the study period is after the anti-corruption campaign. Standard errors are clustered at the district level. See discussion in Section 5.

Table A.18: The effect of university expansion on corruption perceptions and bribe payments, before and after the anti-corruption campaign

	Conditional			Unconditional	
	(1)	(2)	(3)	(4)	(5)
	Perception	Procedure	Service	Procedure	Service
University Expansion	0.0367 (0.0280)	0.0016 (0.0073)	-0.0031 (0.0139)	0.0001 (0.0022)	-0.0007 (0.0029)
University Expansion $\times$ Post 2016	0.0262 (0.0354)	0.0018 (0.0090)	0.0055 (0.0153)	0.0015 (0.0025)	-0.0000 (0.0039)
Observations	163,739	100,431	85,146	171,159	171,159
Cluster No.	320	320	320	320	320
Adjusted R2	0.066	0.018	0.024	0.015	0.028
Mean of Never Treated	-0.026	0.105	0.092	0.023	0.028
District FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Age FE	✓	✓	✓	✓	✓

Note: Appendix Table A.18 shows the effect of the university expansion, before and after the start of the 2016 anti-corruption campaign. The outcome variable ‘Perception’ in Column 1 is a summary index of corruption perceptions, the same as that in Appendix Table A.2. The outcome variables in columns 2 to 5 are indicator variables for having paid a bribe for a government procedure and service, these are the same as in Appendix Table A.3. We combine the age cohort dummies in Equation 1 into an independent variable “University Expansion” which equals 1 if the survey participant was aged 22 or below at the time when the university opened for the individual-level outcome in column 1, and 27 or below for the household-level outcomes in columns 2 to 5. We further interact this indicator variable and fixed effects with the indicator variable, which equals 1 if the study period is after the anti-corruption campaign. Standard errors are clustered at the district level. See discussion in Section 5.